

Research Papers

Techno-economic analysis of grid-integrated PV/wind and storage system for electricity reliability enhancement in the industrial sector in Niger Republic

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ABSTRACT

The unreliability of electricity is a major challenge in many developing countries. In handling this problem, renewable energy resources must be integrated into the existing power grid. Niger, a developing country in West Africa, faces daily electricity shortages, which require the use of diesel generators to ensure operational continuity. However, such a solution poses economic and environmental challenges. This work analyzed the feasibility of integrating photovoltaic (PV)/wind power systems into existing unreliable grid/diesel generator systems to supply industrial critical loads of two selected industries, including ORIBA JUS and AGHAZARMADRAN in Niamey and Agadez regions, respectively. HOMER Pro software was utilized as an optimization tool to perform the techno-economic analysis of the grid-integrated PV/wind. The analysis considered various data, including critical load profile, yearly grid outage duration and frequency, and meteorological data. The most cost-effective system was compared with five different configurations, including grid/diesel (base case because it is the pre-existing system), grid/diesel/PV, grid/diesel/PV/battery, grid/diesel/PV/battery/wind, and grid/diesel/PV/wind, to select the optimal configuration for each region. The study found that the hybrid grid/diesel/PV/wind system is the most cost-effective of all the various configurations for ORIBA JUS in terms of the Net Present Cost (NPC) and Levelized Cost of Energy (LCOE). This system has an NPC of \$1,831,094.00 and an LCOE of 0.082 \$/kWh, which is about \$0.03 lower than the base case. Moreover, the LCOE of this configuration is \$0.011 and \$0.068 lower than the grid electricity price during the non-peak and peak demand periods, respectively. The lowest cost system for AGHAZARMADRAN in the Agadez region is grid/diesel/PV/wind, with an NPC of \$206,927.60 and an LCOE of 0.055 \$/kWh. The NPC of this system is 55 % lower than the grid/diesel system, while the LCOE is approximately \$0.07 lower than the LCOE of grid/diesel, representing over 50 % reduction. Comparison to the grid price, shows that the LCOE of grid/diesel/PV/wind is \$0.095 and \$0.04 lower than the rate during the peak and the low demands, respectively, thus representing a reduction in LCOE of about 63 % and 41 % for peak and low-demand grid prices in that order. In addition, integrating renewable energy significantly reduced fuel consumption by 2474 l/year for ORIBA JUS and 1139.00 l/year for AGHAZARMADRAN, while the CO₂ emissions were reduced by 33 % for ORIBA JUS and 76 % for AGHAZARMADRAN. It was concluded that the integration of PV and wind systems into the present grid and diesel systems in Niger Republic, is economically and environmentally viable.

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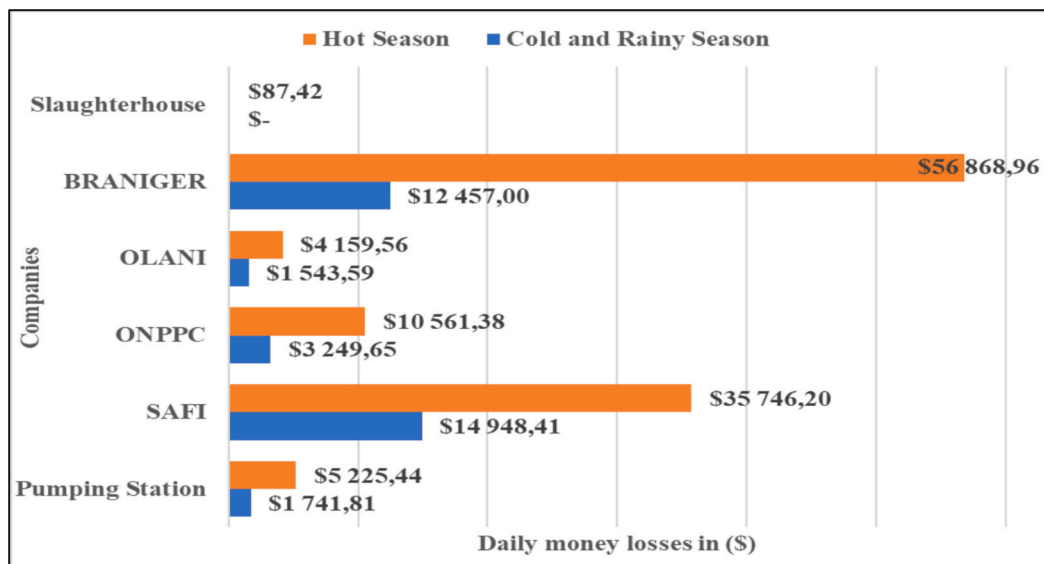


Fig. 1. Estimated daily financial losses linked to power cuts in some large companies in Niger. Source: Niandou, 2021.

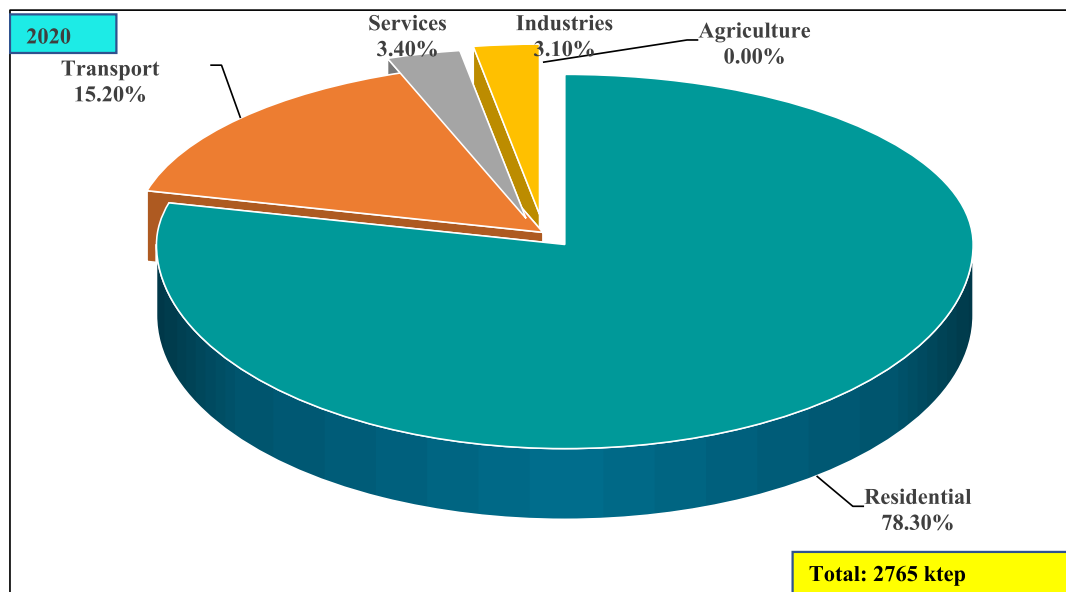


Fig. 2. Energy consumption per sector. Source: SIE-Niger, 2021

1. Introduction

Energy is a crucial factor for the socio-economic development of a country. It exists in different forms, with electricity being the highest grade. Electricity has become unavoidable in our daily activities, and it is of paramount importance to the economic well-being and social condition of a country. It is also worth noting that industrial, commercial, and residential activities revolve around electricity; hence, any unreliability in the electricity delivery to these sectors will considerably affect the country's socio-economic development. According to Azerfegn et al. [1], most developing countries face power uncertainties and shortages due to a highly unreliable electricity supply system.

Technically, power unreliability manifests in two main ways: (i) via a blackout, which refers to the total unavailability of power for the short or long term, and (ii) through electricity brownouts, which entail a drop in electric voltage. These two phenomena have several negative impacts on household and industrial appliances as well as the business climate of a country. For instance, brownouts can cause irreparable damage to

electromagnetic and high voltage appliances such as electric motors, refrigerators, air conditioners, etc. The outages will lead to the total interruption of activities of a firm, which can cause negative socio-economic and environmental impacts, as companies and households will have to rely on diesel generators as backup during these periods of power outages; running on diesel generator has been found to be costly and also serves as a major pollutant [2].

Generally, a country's economic development is defined by its level of industrialization [3]. In developing countries, electricity unreliability presents a significant obstacle to industrial development. Issues of grid outages thus affect the industrial sector heavily by causing huge financial and material losses. For an industry to remain competitive in the market, a reliable [4] and affordable supply of electricity is essential [5]. However, unreliable electricity impacts the growth of the industrial sector in several ways. In the work of Moussa and Malcolm [2], three ways were identified in which electricity unreliability can affect economic transformation, including (i) high production costs for an existing company while creating uncertainties for the establishment of new

Table 1
Review of previous studies.

References	Location	Year	Mode of connection	Systems architecture	Capacity	Mode of application	Sensitivity analysis on:				Comparison with other climatic zones
							Grid price	Diesel price	Load demand	Duration of grid outages	
[1]	Ethiopia	2020	On-grid	- GRID/DG - GRID/DG/PV/BESS - GRID/PV/BESS - GRID/DG/PV/WECS/BESS	44.64 kW 41.66 kW 36.09 kW	Industrial	Yes	No	Yes	Yes	Yes
[8]	Niger	2020	On-grid	GRID/PV	7 MW	Community	No	No	No	No	No
[10]	Pakistan	2024	On-grid	- GRID-DG - On-grid/BG - GRID/PV/BESS	8264.71 kW	Industrial	No	No	No	No	No
[11]	Indonesia	2016	On-grid	GRID/PV/WECS/BESS	970.63 kW	Industrial	Yes	No	No	No	No
[13]	Algeria	2011	On-grid	- GRID - GRID/PV/WECS	95 MW	Community	No	No	No	No	No
[14]	Pakistan	2022	On-grid	GRID/PV		SME	No	No	No	No	Yes
[15]	Sierra Leone	2021	On-grid	- DG - PV/WECS/DG/BESS - PV/WECS/BESS - GRID/PV	1900 kW 1 MW	Rural electrification	No	No	No	No	No
[16]	Saudi Arabia	2017	On-grid	GRID/PV	1 MW	Community	No	No	No	No	No
[17]	Nigeria	2019	Off-grid	PV/WECS/DG/BESS	64.29 kW 2.34 kW	Community	No	Yes	No	No	No
[18]	India	2020	Off-grid	Diesel, PV/diesel/battery, wind/diesel, and PV/wind/diesel/battery	21.17 kW	Community	No	No	No	No	No
[19]	Pakistan	2023	Off-grid	PV/BESS	9.05 kW	Community	No	No	No	No	No
[20]	Libya	2023	off-grid	- PV/WECS/BESS - PV/BESS - WECS/BESS	43.45 kW	Industrial	No	No	No	No	No
[21]	Ghana	2024	Off-grid	PV/DG		Industrial + community	No	No	No	No	No

companies, thus making the country less attractive for private investment, (ii) effect on the productivity of existing firms causing important revenue losses, which can consequently compel companies to shut down production or relocate to an area with a reliable electricity supply, and (iii) effect on the welfare and quality of life of people, which, in turn, can negatively impact the quality of labour in the industry.

Niger, a developing country in West Africa, faces the daily challenge of electricity shortages due to poor services. These unexpected outages occur throughout the year, regardless of the season. The root cause of this issue is the country's dependence on Nigeria for energy imports, as well as insufficient/limited coverage of network infrastructure and obsolete grid networks [6]. Like other sub-Saharan African countries, the unreliability of the grid represents a great obstacle to the industrial development of Niger. A company can experience about 18 power outages per month [3] for around 8 h [7]. These power outages require the company to use diesel generators to ensure seamless operations [6]. These alternative backup solutions pose economic and environmental impacts on industries through materials and financial losses (Fig. 1). For instance, long power outages can cause huge financial losses by interrupting the production chain, while brownouts can cause material losses to sensitive equipment like electric motors. A means of mitigating this issue of electricity unreliability in the context of global warming and energy consumption growth requires integrating renewable energy (RE) resources into the country's energy mix. Niger is naturally endowed with vast RE resources, especially solar energy, with average solar irradiation ranging from 5 to 7 kWh/m² and an average sunshine duration of about 8 h [8]. The wind potential is also enormous, particularly in the northern part of the country [9]. Despite the large deposit of these renewable resources, they only contribute 0.1 % to the energy supply. However, in

recent years, there has been an increased focus on the integration of renewable energy into the electricity grid; as part of efforts to move towards renewable energies, the government has set a target to increase the share of renewables in the energy mix to 30 % by 2030 [9].

The residential sector is the largest primary source of energy consumption in 2020 in Niger, accounting for 78.3 %, and which can be attributed to the country's population relying on biomass being the predominant energy resource. The transport and service sectors followed with 15.2 % and 3.4 %, respectively. However, the industrial sector represented only 3.1 % of the total energy consumption in 2020; this is relatively low compared to other sectors (Fig. 2), and found to be associated with the limited industrial activities in the country. The integration of renewable energy into the electricity grid has been at the center of research in recent years. In particular, the use of photovoltaic (PV) and wind power systems has gained significant attention due to their potential in enhancing the reliability of electricity supply, and most important for industries in locations where access to reliable electricity is limited.

Grid-integrated renewable resources refer to incorporating RE resources into the existing electrical grid infrastructure. This integration allows generated electricity from RE sources, such as solar and wind etc., to be seamlessly connected to the power grid. The goal is to enhance the overall reliability, sustainability, and efficiency of the electrical grid while reducing reliance on traditional fossil fuels. The rising energy demand, unreliable power supply, depletion of fossil fuel, and the environmental challenges associated with climate change, have pushed researchers to carry out several studies on the integration of RE into the existing grid. The integration of PV/wind in the grid can significantly contribute to achieving a reliable, cost-effective, and environmentally

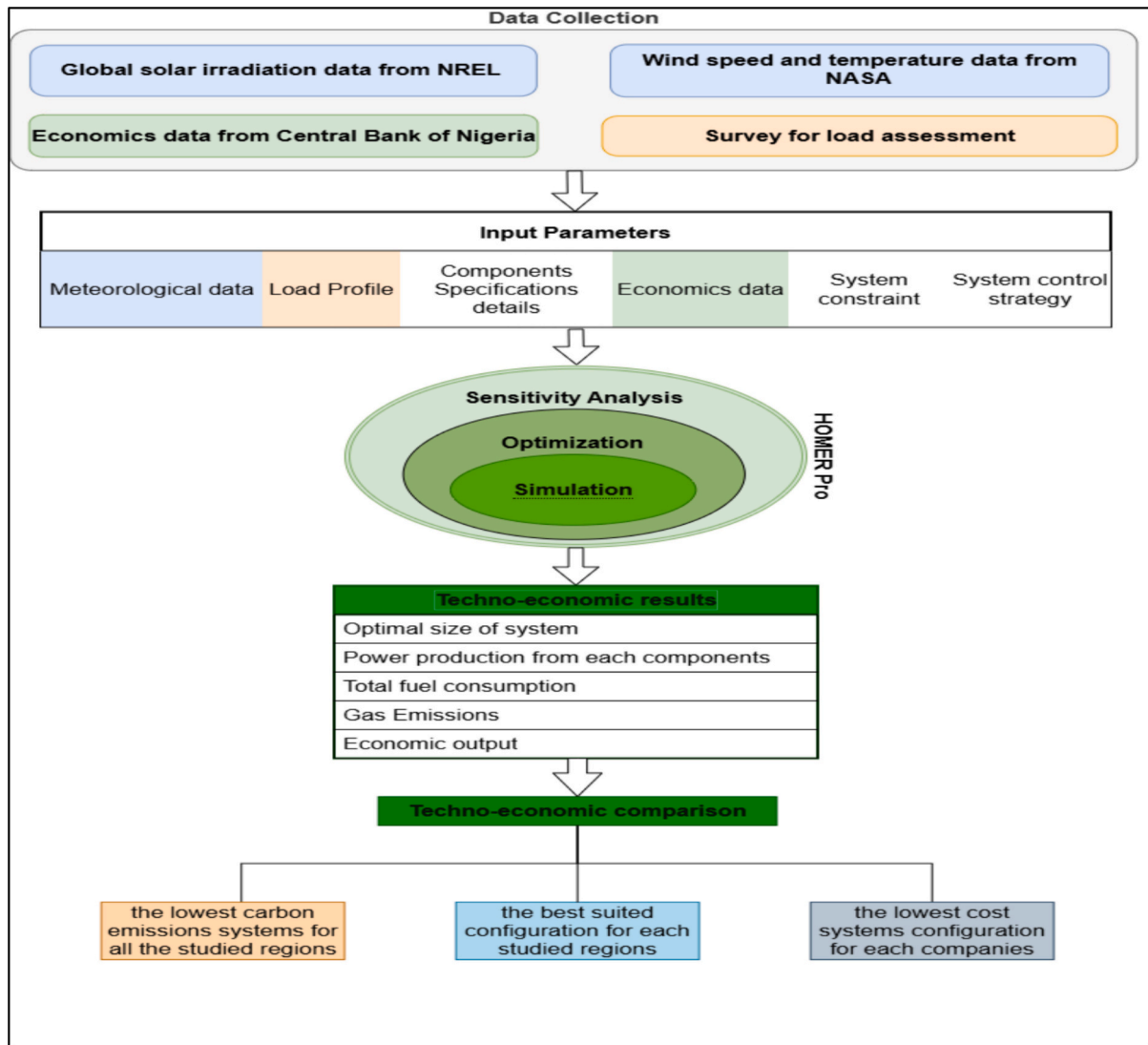


Fig. 3. Flow chart for the study.

friendly power system. Many previous studies have highlighted the importance of cost and emission reductions associated with PV/wind integration.

The techno-economic analysis of grid-integrated PV/wind systems evaluates the technical performance, economic feasibility, and environmental benefits of these systems under different scenarios and assumptions. Various studies have thus examined the techno-economic feasibility of grid-integrated hybrid renewable energy systems (HRES) using different optimization methods. The work of Mumtaz et al. [10] examined the optimal hybrid energy system (HES) for the industrial sector of Pakistan. The results revealed that the integration of renewable energy resources into conventional power systems can significantly reduce the Net Present Cost (NPC), Levelized Cost of Energy (LCOE), and emissions. In a study conducted by Azerefeqn et al. [1] using HOMER software, the techno-economic feasibility of renewable energy systems was investigated in the context of three industrial parks in Ethiopia, and they found that the grid/diesel/PV/battery systems are technically, economically, and environmentally feasible for all the three selected regions. The work by Suamir [11] analyzed the techno-economic benefits of grid-integrated PV/wind systems in Bali, Indonesia, and they concluded that grid-integrated PV/wind can significantly reduce electricity costs and enhance energy security. Similarly, a PV/wind/diesel hybrid system was conducted for rural electrification in Kaduna state,

northern Nigeria, using HOMER as an optimization and modeling tool in the work of Sofimieari et al. [12]. The authors found that the system could provide a reliable and sustainable electricity supply while reducing the dependance on fossil fuels and the associated costs and environmental impact. A study conducted by Saheb-Koussa et al. [13], examined a substantial hybrid power system that combined solar and wind energy with a capacity of 95 MW. The work focused on the environmental and economic implications of the system by utilizing HOMER simulation software to compare grid-connected photovoltaic/wind configuration with a grid-only option, while taking into account the LCOE, NPC, emissions of carbon-dioxide, sulfur-dioxide, and nitrogen oxides; it was concluded in the work that the PV/wind configuration was more advantageous. A study was conducted in four locations in Pakistan on grid-connected solar PV power plants for small-scale industries using RETScreen Expert software to highlight the technical, financial, and environmental viability of the project [14]. The work of Conteh et al. [15] adopted a genetic algorithm to analyze the techno-economic feasibility of integrating a hybrid system in the grid for reliable and sustainable rural electrification in Sierra Leone. The study aimed at lowering the loss of power supply probability (LPSP) and the cost of energy (COE); among the three configurations considered in their work, the PV/wind/diesel/battery was found to be the most cost-effective.

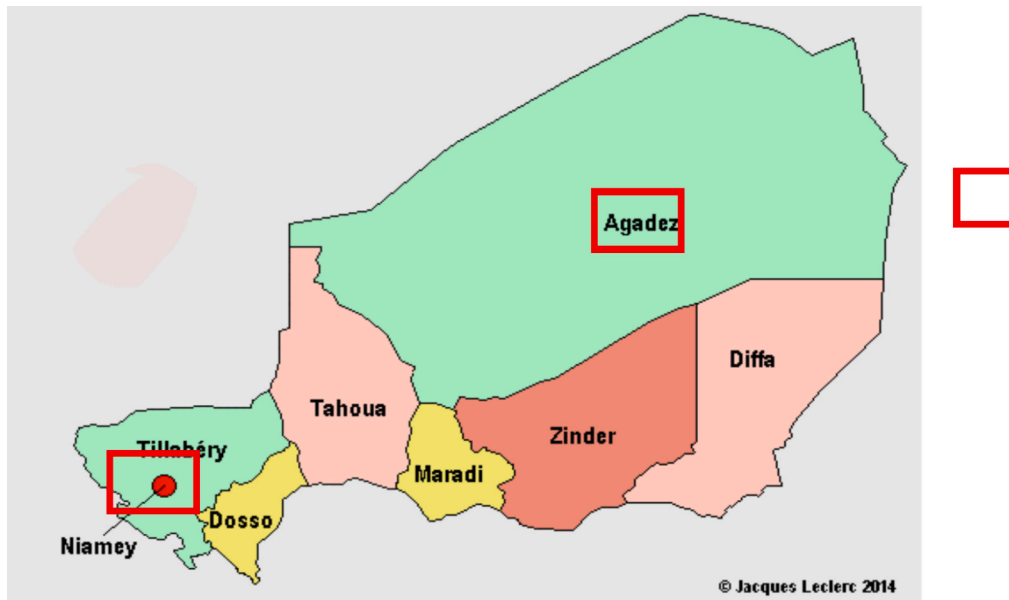


Fig. 4. Location of the selected sites sources. Niger: carte (ulaval.ca).

1.1. Research contribution

The techno-economic analysis of grid-integrated HRES has been widely studied with focus on rural applications and on specific sites as summarized in Table 1. It is worth noting that these studies neglected the variations of essential variables such as load demand and diesel costs, durations of grid outages, etc. This research thus aims to design a reliable, cost-effective, and environmentally sustainable HRES for critical industrial loads using Homer Pro software for two different companies located in different climatic regions of Niger Republic. It considers various design parameters using climatic and economic data, and with rigorous sensitivity analysis on key variables to enhance simulation accuracy. This work thereby focuses on:

- Developing a detailed daily load profiles for two different industries through comprehensive data collection with data loggers.
- Evaluating five distinct hybrid system configurations for each company to determine the most suitable solution, based on their operational needs and regional conditions.
- Conducting a cost-effective analysis needed to identify the most economically viable systems.
- Performing sensitivity analysis on critical factors such as grid selling prices, diesel costs, grid outages, wind speed, solar radiation, and load demand, while at the same time assessing their impact on NPC and COE over the lifetime of the project.
- Evaluating the environmental impacts of each hybrid system.

This analysis seeks to enhance the industrial sector of Niger, by proposing a reliable, and cost-effective hybrid systems needed to achieve electricity dependance. It will also serve as a guide to business owners, renewable energy innovators, and government organizations towards the adoption of renewable solutions, needed for electricity reliability across the different regions of Niger republic.

2. Methodology

In this work, the techno-economic analysis of integrating PV and wind systems for the industrial sector (having unreliable grid and diesel system) in two different regions of Niger was conducted using HOMER software. HOMER software is a robust resource for creating and refining Hybrid Renewable Energy Systems (HRES) by simulating diverse energy

sources. It determines the most efficient and cost-effective combination of resources, system size, and storage capacity. A techno-economic comparison between the two regions is also performed. This study requires various types of data input; hence, many surveys were performed to capture the data. The dataset includes energy consumption variation, information on backup generators of the industries, technical and economic details of components, grid reliability data, and meteorological data for each of the selected regions. The flow chart of the study is shown in Fig. 3.

2.1. Study area

To conduct this research, two cities namely Niamey and Agadez, located in distinct climatic areas of Niger, were selected. Niamey is in the west of the country in the Sahelo-Sudanian climatic zone, and Agadez is in the arid zone in the northern part (Fig. 4).

2.1.1. The region of Niamey

Niamey, the capital of Niger, is located in the extreme west of the country. It has an area of 552.27 km², with a population of 1,407,635 inhabitants in 2022 [22] and is divided into five administrative communities. The region benefits from a Sahelo-Sudanian climate characterized by a short rainy season (June to September) and a long dry season (October to May). The selection of Niamey for this study was based on its status as the capital; most of the industries in Niger are located in Niamey and are powered by electricity imported from Nigeria, in addition to diesel generators.

2.1.2. The region of Agadez

Agadez is located in the northern zone with a desert climate. It has an area of 667,799 km², which represents 52.6 % of the country's area. Agadez is divided administratively into six communities. The sub-desert climate in the tropical region is extremely dry, with minimal rainfall and high evaporation levels that worsen the aridity of the climate. The region has a sharp thermal gradient characterized by large thermal amplitudes (maximum 42 °C and minimum – 2 °C), and prevailing winds from the North/North East (wind observations >25 km/h at Bilma on 160 days/annum). Despite the abundant renewable energy resources in the region, it is supplied by a single coal power plant, which always experiences frequent power cuts in addition to pollution.

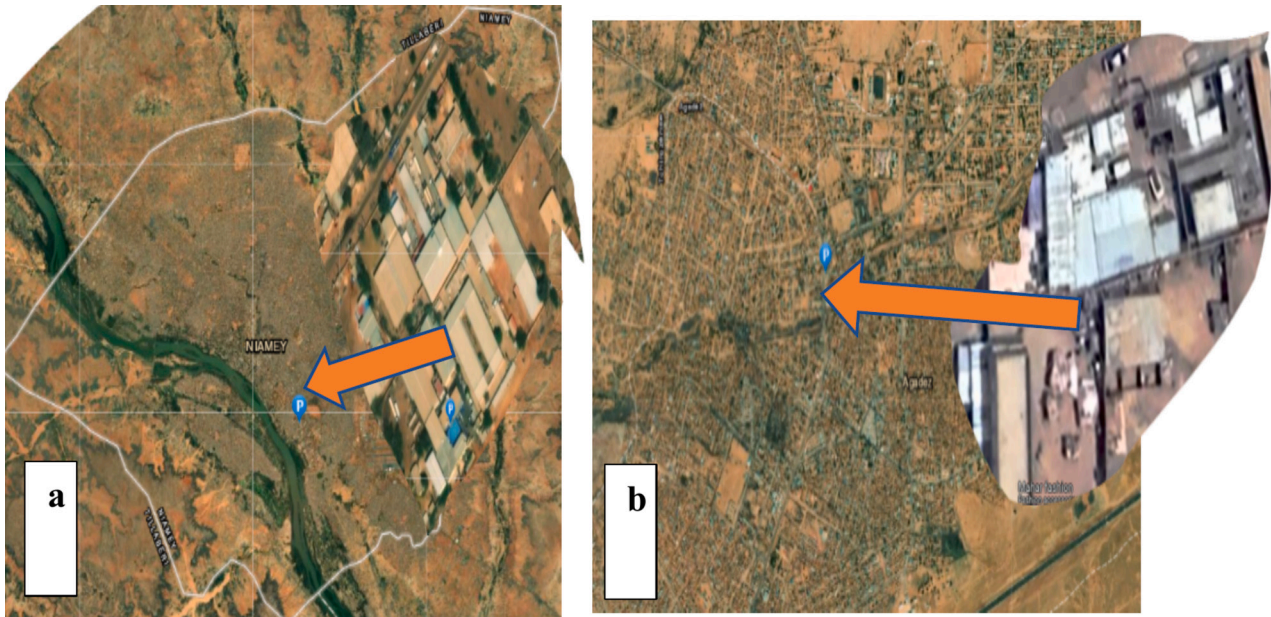


Fig. 5. a) ORIBA JUS, b) AGHAZARMADRAN. Source: NASA.

Table 2
Geographical coordinates of selected industries.

Region	Company	Coordinates		
		Latitude (N)	Longitude (E)	Elevation (m)
Niamey	ORIBA JUS	13.4928°	2.1327°	207
Agadez	AGHAZARMADRAN	16.9774°	7.9838°	520

Table 3
Global horizontal and on-tilt surface solar radiation in Niamey and Agadez.

	Niamey		Agadez	
	Average daily GHI (kWh/m ² /day)	Clearness index	Average daily GHI (kWh/m ² /day)	Clearness index
Jan	5.161	0.603	5.572	0.696
Feb	5.34	0.603	6.134	0.686
Mar	4.824	0.584	6.675	0.673
Apr	4.416	0.564	7.164	0.679
May	3.727	0.537	7.147	0.666
Jun	3.239	0.519	6.823	0.636
Jul	3.132	0.497	6.795	0.636
Aug	3.421	0.483	6.459	0.614
Sep	4.176	0.538	6.488	0.646
Oct	4.898	0.59	6.246	0.681
Nov	5.228	0.646	5.782	0.703
Dec	5.256	0.608	5.320	0.689
Mean	5.94	0.56	6.38	0.67

2.2. Selected companies for the case study

In each town, one company was selected for the case study (Fig. 5). Table 2 presents the name and geographical coordinates of each of the selected companies. ORIBA JUS company is situated in the industrial zone with an area of 3600 m² and specializes in the production of juice called ORIBA. The power subscription from the national electricity utility company is 2000 kW. The company is equipped with two diesel generators as backup, including 800kVA and 400kVA. This study is concerned with only the critical load for which the 400kVA generator is utilized as the backup. AGHAZARMADRAN is a bakery company that specializes in pastries and restaurants. It has a surface area of 800 m² and is equipped with a single generator having a capacity of 45 kVA as a

backup. The capacity subscription on the national electricity utility is 30 kW.

2.3. Data collection

Meteorological data, including solar radiation, wind speed, and temperature, for 10 years were collected from the National Meteorological Direction at Niamey Aero station for each town. Grid reliability data needed for grid modeling was collected from the national electricity utility companies. To obtain the load profile, a survey was conducted on one industry in each of the selected towns to capture critical load profile information and the existing backup generator details. ORIBA-Jus company and AGHAZARMADRAN Boulangerie-Pâtisserie-Restaurant were selected in Niamey and Agadez, respectively.

2.3.1. Resource assessment

A 10-year daily solar radiation dataset from 01/01/1996 to 12/31/2007 and daily wind potential and temperature from 01/01/2014 to 12/31/2023 were collected from the National Meteorological Direction, Niamey Aero station, for Niamey. In Agadez, 10-year daily meteorological data (solar radiation, wind speed, and maximum temperature) were obtained from NASA. The 10-year daily data were cleaned up and merged to get a monthly average data using the Pandas module on VS code for the solar radiation (see Supplementary material).

2.3.2. Solar resource

Table 3 shows the daily solar radiation and clearness index of Niamey and Agadez. In Niamey, the annual average global horizontal solar radiation is 5.94 kWh/m²/day. The highest and lowest global solar radiation occurs in March and December, respectively, with values of 6.55 kWh/m²/day and 5.34 kWh/m²/day. The clearness index, which measures atmospheric transparency, has a mean of 0.56 and ranges from 0.48 to 0.64 in August and November, respectively. In Agadez, the annual average global horizontal solar radiation is 6.38 kWh/m²/day, with the maximum and minimum values observed in April (7.164 kWh/m²/day) and December (5.320 kWh/m²/day), respectively. The clearness index in Agadez ranges from 0.614 to 0.696, with an average daily value of 0.67. Fig. 6 shows the plotted solar radiation and clearness index of Niamey and Agadez.

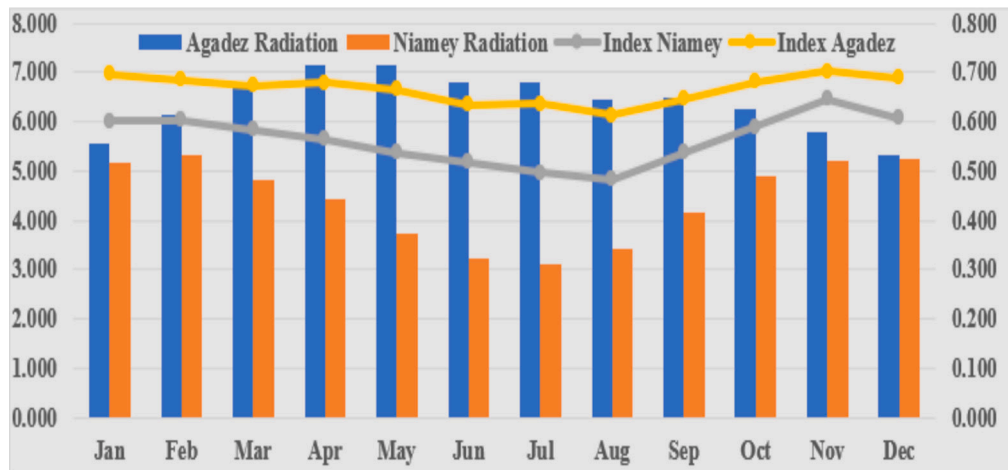


Fig. 6. Solar radiation and clearness index for Niamey and Agadez in kWh/m²/day.

Table 4

Average wind speed and power density in Niamey and Agadez at 10 m.

	Niamey	Agadez
	Average daily wind speed (m/s)	Average daily wind speed (m/s)
Jan	3.616	5.531
Feb	3.597	5.555
Mar	3.265	4.875
Apr	3.268	4.459
May	3.758	4.214
Jun	3.97	3.611
Jul	3.625	3.602
Aug	3.11	3.501
Sep	2.891	3.457
Oct	2.624	4.605
Nov	2.537	4.843
Dec	3.388	5.551
Mean	3.304	4.484

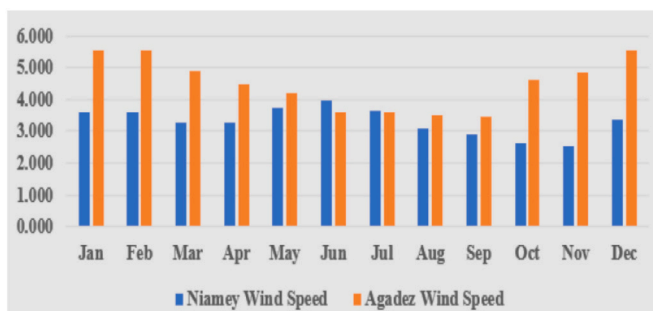


Fig. 7. Average monthly wind speed at 10 m in Niamey and Agadez.

2.3.3. Wind resource

A 10-year daily wind speed data were analyzed and cleaned using Jupyter Notebook on VS Code and Microsoft Excel (see Supplementary material). In this study, analysis of the wind resource consists of determining the power density and Weibull probability function for each location. Table 4 and Fig. 7 display the monthly average wind speeds and the power density at a height of 10 m for Niamey and Agadez. In Niamey, the annual average wind speed at 10 m is 3.304 m/s. The highest and lowest wind speeds observed in June and November were 3.970 m/s and 2.537 m/s, respectively. For Agadez, the annual average wind speed at 10 m is 4.484 m/s, with the maximum and minimum values observed in February (5.56 m/s) and September (3.46 m/s), respectively.

2.3.4. Temperature

The efficiency of PV modules is significantly affected by the ambient air temperature of the location of the facility, which makes it a key factor in evaluating the performance of PV systems. To account for the impact of temperature in this study, we gathered 10-year daily air temperature data. This data was curated and uploaded into HOMER. The maximum monthly average temperature for Niamey and Agadez is illustrated in Fig. 8 and Table 5. In Niamey, the average temperature fluctuates between 32.64 °C (January) and 41.96 °C (April), with an overall annual average of 36.74 °C, while in Agadez, it varies from 28.00 °C (January) to 41.82 °C (May). The annual average is 36.07 °C.

2.3.5. Load assessment of the selected industries

Metrol ME435 Energy Analyzer was utilized to capture the load profile of each company. The data logger was installed at the departure cable of the power supply and set to capture and log a 5-minute step data for the week. The data logger records various parameters, including the active, reactive, and apparent power and energy for the three phases. It also calculates the sum of the three-phase power and energy. The daily energy demand is considered to be 2004.4 kWh/day for ORIBA JUS and 340.65 kWh/day for AGHAZARMADRAN. The daily load profile of the critical load for ORIBA JUS and AGHAZARMADRAN is shown in Fig. 9. It can be seen that the load profile for ORIBA JUS has a dip during the afternoon time, particularly between 10:00 am and 4:00 pm. This is due, on the one hand, to the break time during which most of the equipment is shut down and, on the other hand, to the fact that most of the production occurs in the evening during the low demand period, which is from 00:00 am to 10:00 am and from 4:00 pm to 11:00 pm; during this period electricity is also cheaper. The load profile for AGHAZARMADRAN shows that the lowest consumption is observed between 00:00 am and 6:00 am. This is because the company stops its activities by 11:00 pm. From 6:00 am, the consumption increases gradually to reach its peak at 7:00 am.

2.4. System description and specifications of components

Fig. 10 shows the schematic diagram of the system configuration. The base case system consists of grid/diesel, where during a grid outage, diesel supplies the total load of the company. The proposed system consists of integrating PV, wind turbine, battery, and inverter to support the diesel generator in supplying critical load during grid outages. Fig. 11 shows the diagrams of the grid-connected system in HOMER Pro for (a) Niamey and (b) Agadez.

2.4.1. Grid outages and tariffs

To model the grid reliability, the tariff mode, the average frequency

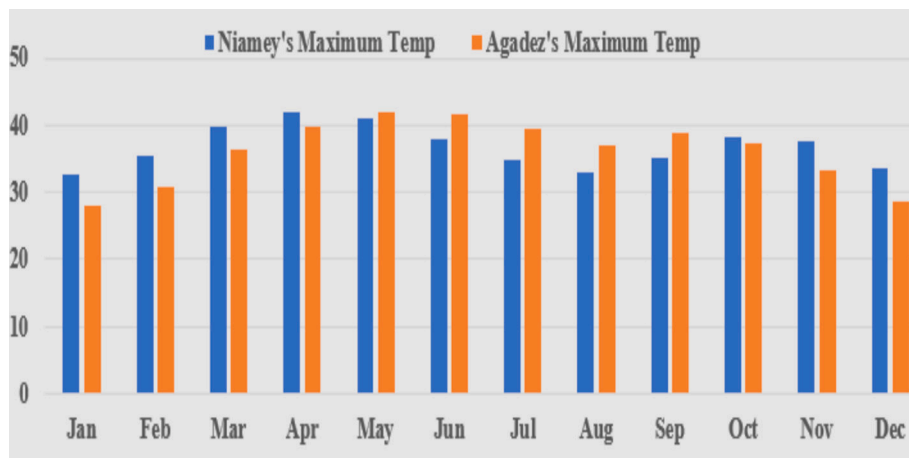


Fig. 8. Monthly average maximum temperature in Niamey and Agadez.

Table 5
Monthly average maximum temperature in Niamey and Agadez.

Month	Niamey	Agadez
Jan	32.64	28
Feb	35.58	30.93
Mar	39.76	36.26
Apr	41.96	39.71
May	40.96	41.82
Jun	38.03	41.67
Jul	34.71	39.5
Aug	32.88	36.91
Sep	35.01	39.02
Oct	38.22	37.23
Nov	37.64	33.28
Dec	33.55	28.53
Annual mean	36.74	36.07

of the grid failure, and the average duration of interruption on the power line that supplies each company are collected from the national electricity utility company. For the industrial sector, the electricity tariff

consists of two different patterns, i.e., the peak hour tariff from 10:00 am to 4:59 pm has the tariff as 0.15\$/kWh, and the non-peak hour tariff of 0.093\$/kWh from 5:00 pm to 9:59 pm. The grid outage data for each company is shown in Table 6.

2.4.2. Grid modeling

HOMER can use the data presented in Table 6 to simulate an unreliable power grid. This simulation includes both scheduled and unscheduled outages, which span over one year. HOMER computes the average outage frequency and average repair time to create a model of a random grid outage that may occur over the entire year. It is worth noting that the study only considers unscheduled grid outages. The results of the grid modeling for ORIBA JUS and AGHAZARMADRAN are demonstrated in Figs. 12 and 13, respectively.

2.4.3. Diesel generator

A generator is a self-contained device that provides a portable or stationary source of power. It is the most commonly used method of power generation, particularly in regions where the public electricity

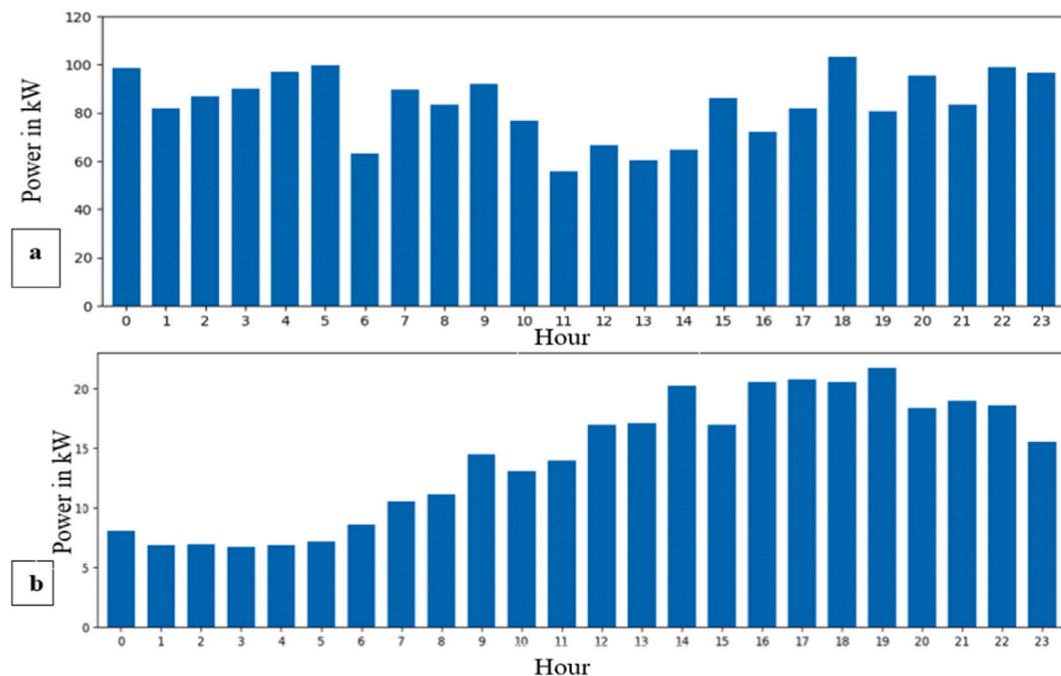


Fig. 9. Daily load profile of considered critical load for (a) ORIBA JUS, and (b) AGHAZARMADRAN.

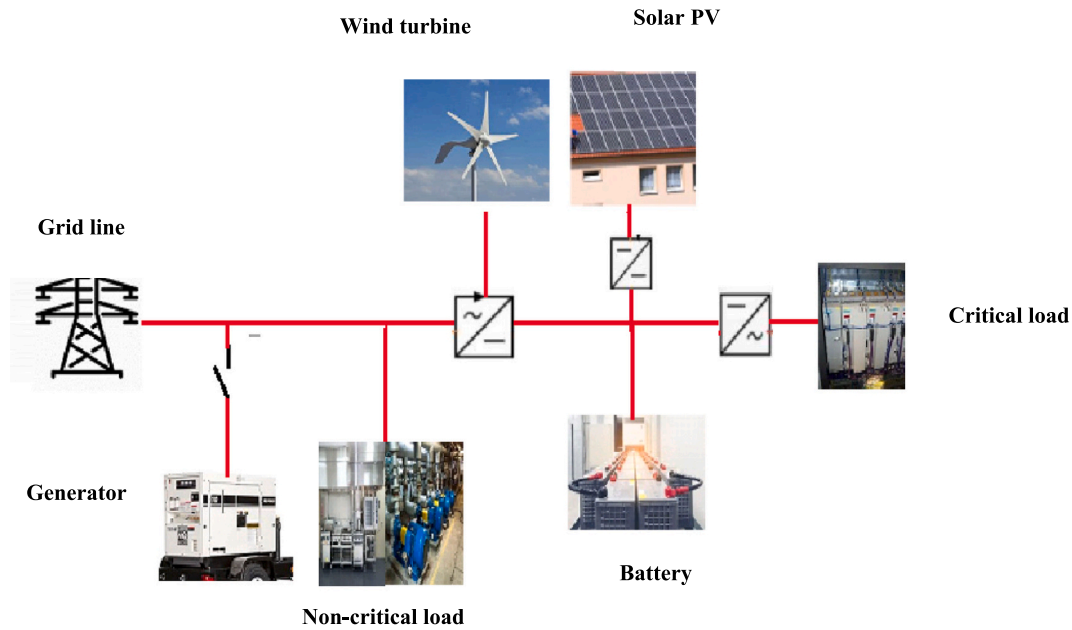


Fig. 10. Schematic diagram of the proposed system.

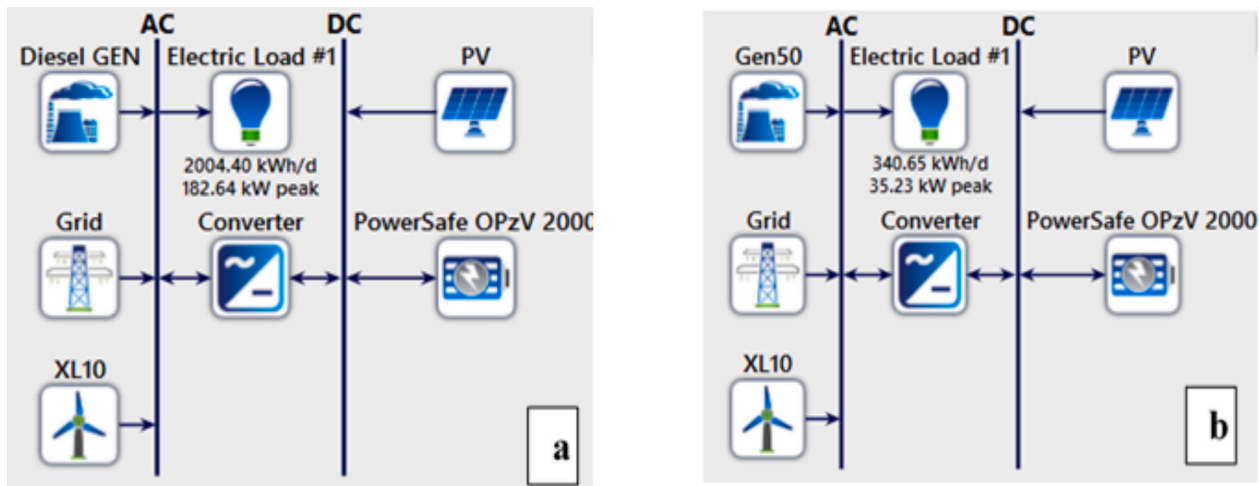


Fig. 11. Diagram of the grid integrated system in HOMER Pro for (a) ORIBA JUS and (b) AGHAZARMADRAN.

Table 6

Grid outage data for each company.

Company	Year	Grid interruption frequency per year	Average interruption duration (hour/year)	Electricity tariff for industry	
				Peak hour	Off-peak hour
ORIBA JUS	2023	247	131.33	\$0.15	\$0.093
AGHAZARMADRAN	2023	74	260.417	\$0.15	\$0.093

network is unreliable or unavailable. Generators consume fuel to produce electricity and, in some cases, heat as a by-product [23,24]. When modeling a generator, several essential parameters need to be taken into account. These include the maximum and minimum electrical power output, the expected lifetime of the generator in hours, and the quantity of fuel that the generator consumes per hour of operation. Additionally, it is important to consider the fuel curve, which serves as a link between fuel consumption and the electrical power generated [23]. By analyzing

these factors, it becomes possible to create an accurate and effective model for the generator that can be used to optimize its performance and ensure its continued operation over time. Eq. (1) links the fuel consumption (units per hour), to the output power P_{gen} of the diesel generator (kW):

$$F = F_0 * Y_{gen} + F_1 * P_{gen} \quad (1)$$

In Eq. (1), F is the fuel consumption in litre/h, F_0 is the intercept coefficient of the generator fuel curve in litre/h/kW rated, F_1 is the fuel curve slope in litre/h/kW output, Y_{gen} is the nominal capacity of the diesel generator in kW, and P_{gen} is the electrical power output of the generator (kW). The technical and economic data of the existing generators for each company are shown in Table 7.

2.4.4. PV array

In HOMER, the photovoltaic (PV) array is modeled as a generator that delivers direct current electrical power output by using Eq. (2). The modeling approach of HOMER considers the various factors that can affect the performance of the PV array, such as solar radiation and

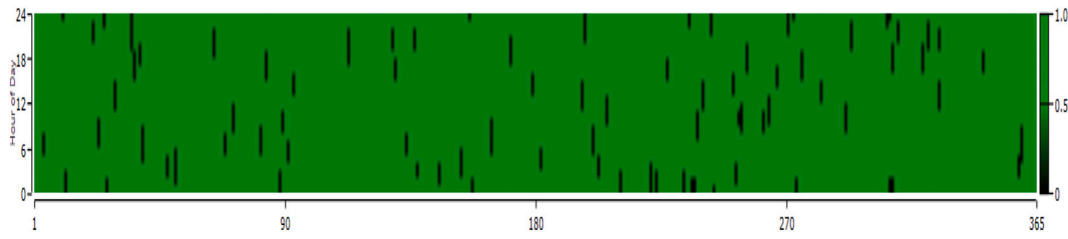


Fig. 12. Random grid failure from HOMER for ORIBA JUS.

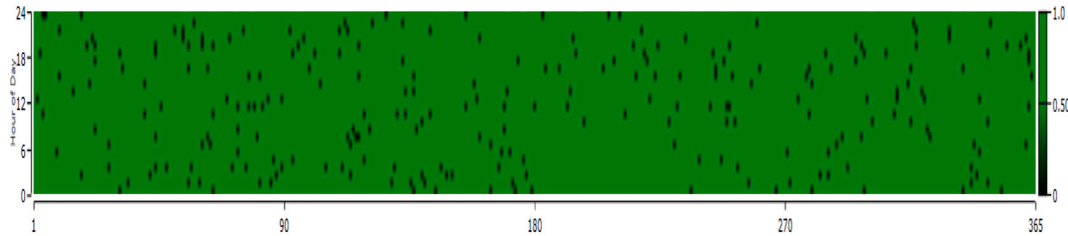


Fig. 13. Random grid failure from HOMER for AGHAZARMADRAN.

Table 7

Technical and economic data of the existing generators.

	ORIBA JUS	AGHAZARMADRAN
Generator type	Synchronous	Synchronous
Rated capacity	320 kW	36 kW
Fuel consumption	100 l/h	7.2 l/h
Capital cost	\$ 0	\$ 0
O & M	0.73 \$/hour	0.1 \$/hour
Replacement cost	\$ 46,080	\$ 6272
Lifetime	50,000 h	15,000 h

Table 8

Characteristics of the PV module.

Parameters	Specification
Module series	TPSh-M2P112SF1W-520 W
Maximum power at STC (P_{max})	520 W
Short circuit current (I_{sc})	9.27 A
Open circuit voltage (V_{oc})	71.71 V
Maximum power current (I_{mp})	8.91 A
Maximum power voltage (V_{mpp})	58.37 V
Module efficiency	17.51 %
Operating temperature	$-40^{\circ}\text{C} + 85^{\circ}\text{C}$
Temperature coefficient of power	$-0.002\%/^{\circ}\text{C}$
Nominal operating cell temperature	47°C
Derating factor	85 %

temperature, among others, to accurately simulate the behavior and output of a PV array in a given system.

$$P_{\text{output}} = f_{pv} * P_{pv} * \frac{G}{G_{\text{STC}}} * [1 + \alpha_p (T_c - T_{\text{STC}})] \quad (2)$$

where, f_{pv} is the derating factor, G is the global solar insolation incident on the surface of the PV (kW/m^2), and P_{pv} is the rated capacity of the PV module. The TPSh-M2P11SF1W-520 PV module is selected in this study. The characteristics of the PV module are shown in Table 8.

2.4.5. Wind turbine

The characteristics of the wind that can be utilized to determine the wind energy potential of a site include speed, direction, continuity, and availability [25]. Evaluating the potential of a site generally consists of determining the wind power density of the site and then the site class. The Battelle-Pacific Northwest Laboratory (PNL) established a scheme of

wind power classification ranging from Class 1 with a power density that is less than or equal to $100 \text{ W}/\text{m}^2$ to Class 7 with a power density that is equal to or above $1000 \text{ W}/\text{m}^2$. To determine the wind potential of a site, researchers use the methods of probability density functions, such as the Weibull and Rayleigh functions. Some authors also use the Wind Atlas Analysis and Application Program (WASP) to evaluate the wind energy potential of a complex region.

The Weibull distribution function is commonly used to determine the wind speed characteristics and to estimate the wind power density because it gives a better fit with measured data than other statistical functions [26,27]. It has two parameters, including the shape parameter (k) and the scale parameter (c). Different methods are used to determine the parameters of Weibull functions. For example, Alsamamra et al. [28] present five numerical methods, including the Empirical Method (EM), Energy Pattern Factor Method (EPFM), Mean Absolute Percentage Error (MPE), Modified Maximum Likelihood Method (MMLM), and Method of Moments (MoM), to determine the Weibull parameters. They found that both EM, which uses the standard deviation, and the MoM were the most accurate in determining the values of the Weibull shape and scale parameters.

The kinetic energy carried by a mass of air moving at a speed v is given by [29]:

$$E_{\text{kinetic}} = \frac{1}{2} * m * v^2 \quad (3)$$

By expressing the mass (m) as a function of A (the rotor area in m^2), ρ the air density in kg/m^3 and v which is the wind speed in m/s , the mechanical power in the upcoming air is given by [29]:

$$P = \frac{1}{2} * \rho * A * v^3 \quad (4)$$

where P is the mechanical power. It is worth noting that the swept area depends on the rotor diameter:

$$A = \frac{\pi * d^2}{4} \quad [\text{m}^2] \text{ for the horizontal-axis wind turbine}$$

$$A = \frac{2}{3} * w_r * h_r \quad [\text{m}^2] \text{ for the vertical axis turbine}$$

The power expressed above cannot be completely captured by the rotor, as the air still has some power downstream, moving at a low speed. Then, the actual power captured by a wind turbine from the wind velocity on a given site is given by the following expression [29]:

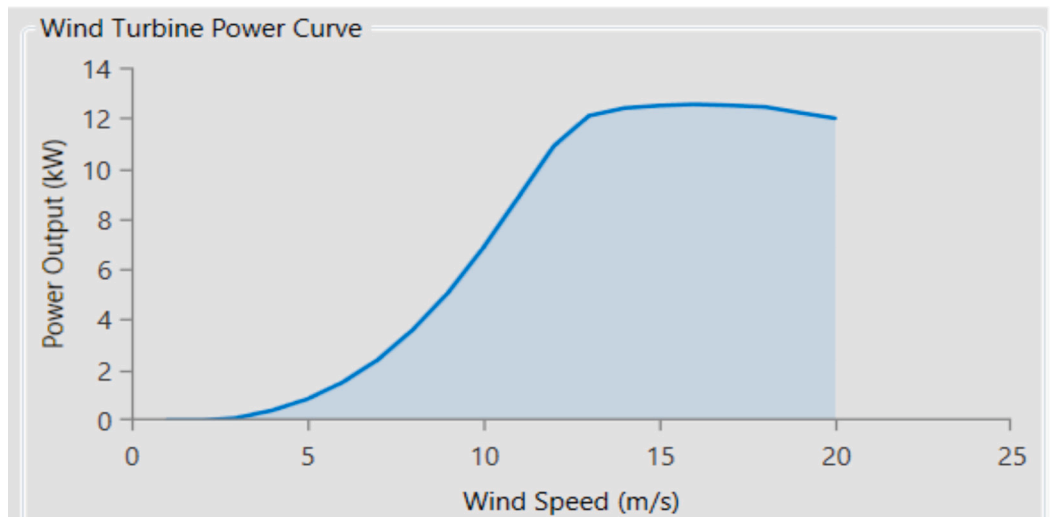


Fig. 14. Power curve for Bergey Excel 10 wind turbine.

Table 9

Wind turbine characteristics.

Parameters	Specification
Model	Bergey Excel 10
Type	3-blade, horizontal axis
Rated power	10 kW
Rotor diameter	7 m
Hub height	24–49 m
Rated speed	11 m/s
Cut-in wind speed	2.5 m/s
Cut-off wind speed	20 m/s
Operating temperature	−40 °C + 60 °C
Weight	545 kg

Table 10

Characteristics of the battery.

Parameters	Specification
Model	PowerSafe OPzV 2000
Maximum capacity	2762.579 Ah
Nominal voltage	2 V
Lifetime	8 years
Lifetime throughout	6906.2 kWh
Roundtrip efficiency	95 %
Maximum charge current	889 A
Maximum discharge current	3067.00 A

Table 11

Economics assumptions.

	PV	Wind turbine	Converter	Battery
Assumed replacement cost as % of the initial cost	80 %	90 %	100 %	90 %
O&M cost per year as % of the initial cost	1 %	2 %	2 %	1 %
Assumed lifetime	25 years	25 years	15 years	8 years
Discount rate	5.25 %			
Inflation rate	6 %			

Table 12

Cost structure of components.

	PV	Wind turbine	Converter	Battery
Unit capacity	0.52 kW	10 kW	1 kW	5.53
Initial cost in \$/unit	\$0.48/W	\$870/kW	\$208/kW	\$125/kWh
Capital cost	\$274	\$11,520	\$270	\$730
Replacement cost	\$0	\$0	\$270	\$660
O&M cost per year	\$2.74	\$230	\$5.41	\$7.30

$$P_0 = \frac{1}{2} \left[\rho * A * \frac{(v + v_0)}{2} \right] * (v^2 - v_0^2) \quad (5)$$

where, P_0 is the mechanical power extracted by the rotor, v and v_0 are the upstream and downstream wind speeds, i.e., the wind speeds at the entrance and exit of the rotor blades. Eq. (5) can be written as:

$$P_0 = \frac{1}{2} * \rho * A * v^3 * \frac{\left(1 + \frac{v_0}{v}\right) \left[1 - \left(\frac{v_0}{v}\right)^2\right]}{2} \quad (6)$$

$C_p = \frac{\left(1 + \frac{v_0}{v}\right) \left[1 - \left(\frac{v_0}{v}\right)^2\right]}{2}$, is the fraction of the upstream wind power extracted by the rotor, and also called the power coefficient. Hence, by replacing C_p , Eq. (6) becomes:

$$P_0 = \frac{1}{2} * \rho * A * v^3 * C_p \quad (7)$$

The assessment of wind potential at a given site is to determine its wind power density or specific power expressed in terms of W/m^2 of rotor area. The specific power is given by [29]:

$$P_{\text{specific}} = \frac{1}{2} * \rho * v^3 [W] \quad (8)$$

The electrical power generated by a wind turbine is given:

$$P_{\text{elec}} = \frac{1}{2} * (C_p * \eta_{gb} * \eta_g) * A * \rho * v^3 \quad (9)$$

The variation in the power production of a wind turbine is given by the power curve function [15]:

$$P_{WT} = \begin{cases} 0 & v < v_{ci} \text{ or } v > v_{co} \\ v^3 * \left(\frac{P_r}{v_r^3 - v_{ci}^3} \right) - P_r * \left(\frac{v_{ci}^3}{v_r^3 - v_{ci}^3} \right) & v_{ci} \leq v \leq v_r \\ P_r & v_r \leq v \leq v_{co} \end{cases} \quad (10)$$

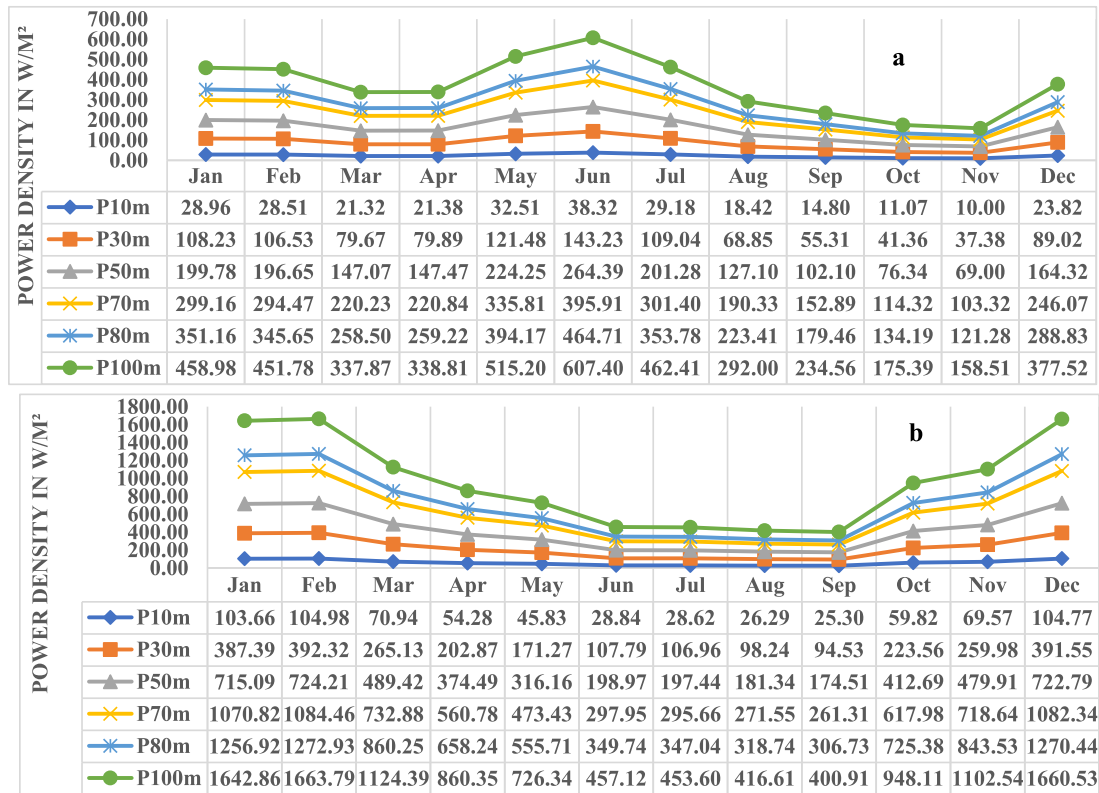


Fig. 15. Power density at different heights in (a) Niamey, and (b) Agadez.

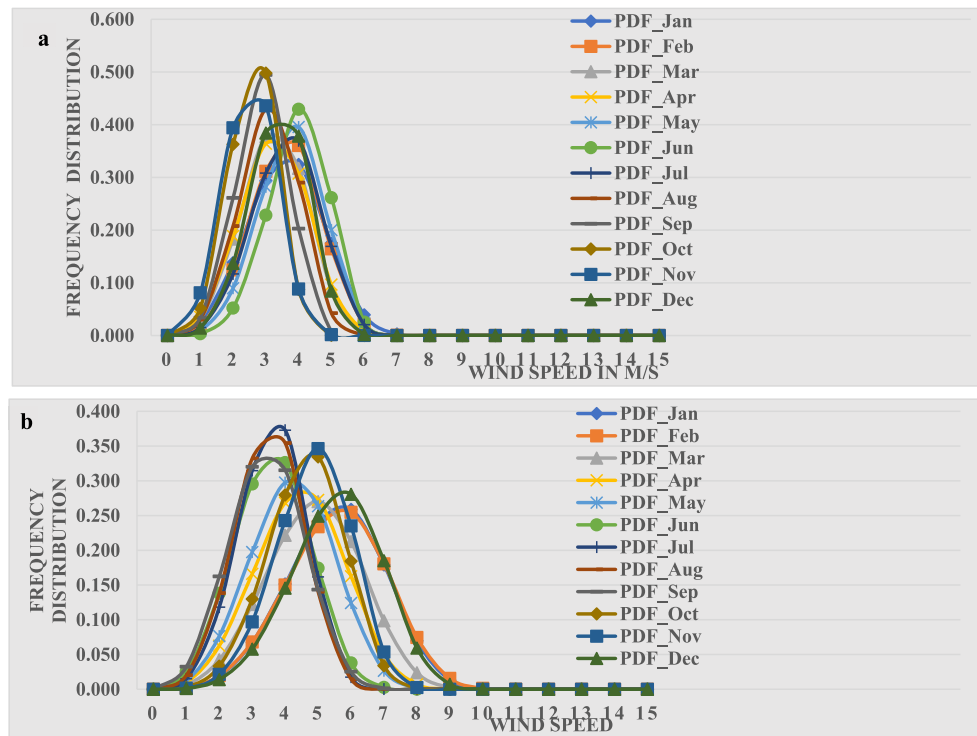


Fig. 16. Monthly probability density function at 10 m in (a) Niamey, and (b) Agadez.

where, P_{WT} is the output power, P_r is the wind turbine-rated power, v , v_{ci} , v_{co} , v_r are respectively the wind speed, cut-in, cut-out, and rated wind speed.

Wind speed is always reported at 10 m altitude, while the hub height

of the wind turbine is always greater than this height for high wind speed. Hence, wind speed needs to be extrapolated to the hub height of the wind turbine. Three main methods are used to extrapolate wind speed from the anemometer height to the wind turbine hub height:

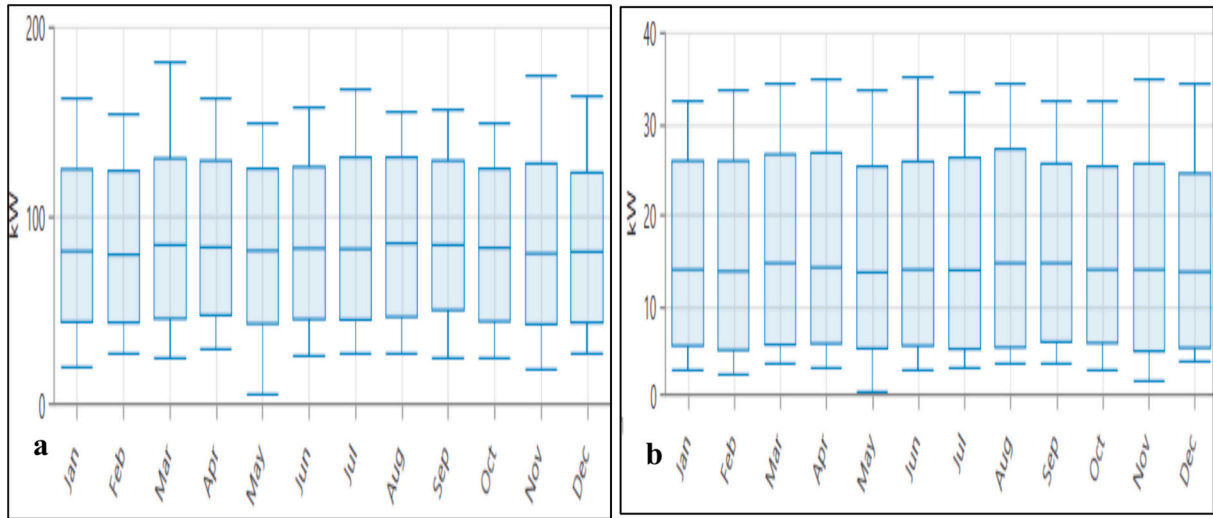


Fig. 17. Seasonal load profile for each company in (a) ORIBAS JUS, and (b) AGHAZARMADRAN.

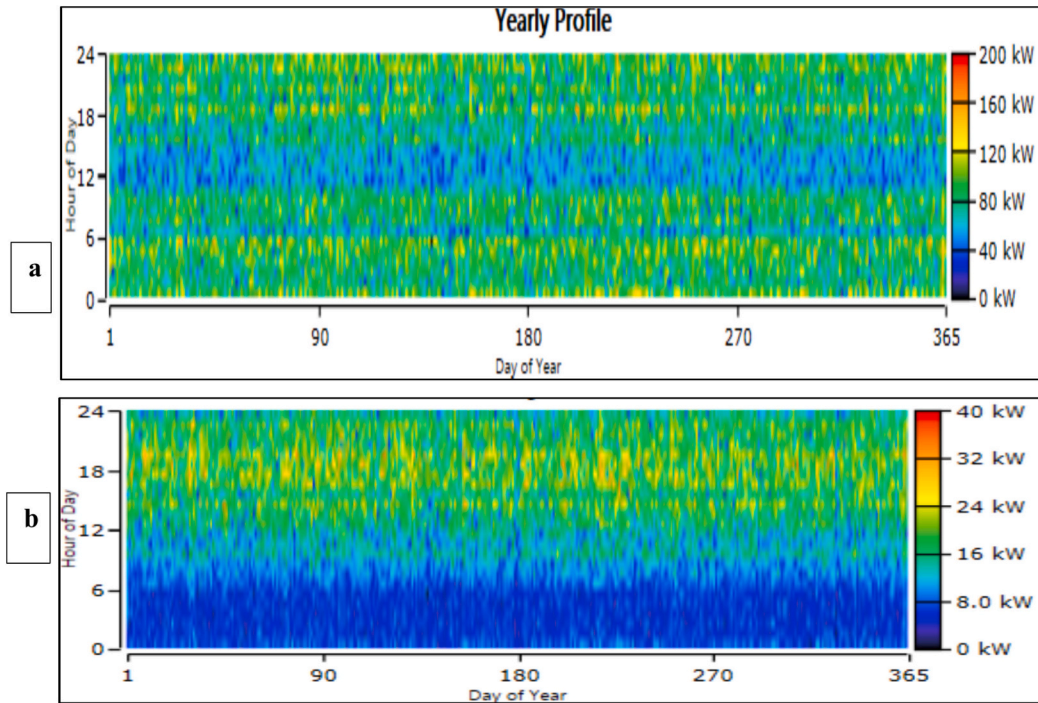


Fig. 18. Yearly critical load profile for each company in (a) ORIBAS JUS, (b) AGHAZARMADRAN.

- The Monin-Obukhov method: This method is useful for calculating wind speed over short periods, usually in minutes, and for averaging readings [30,31].
- Logarithmic law: Also referred to as the logarithmic wind profile method. This method assumes a logarithmic relationship between wind speed and height.
- Power law extrapolation: This method, which is also known as the Hellmann exponential law [30,31] assumes that wind speed increases with height raised to a power. This method is simple and widely used in wind energy applications, but its accuracy depends on the site-specific conditions and the choice of the exponent. In this study power law extrapolation method is used to determine the wind speed at the hub height of the selected wind turbine. This is expressed as:

$$v = v_0 * \left(\frac{h}{h_0} \right)^\alpha \quad (11)$$

where v is the wind speed at height h , v_0 is the wind speed at height h_0 , and α is the friction coefficient or Hellman exponent. A friction coefficient of 0.4 is adopted in this work.

2.4.5.1. Weibull distribution function. The Weibull distribution function is defined as [28,29]:

$$f(v) = \left(\frac{k}{c} \right) \left(\frac{v}{c} \right)^{k-1} \exp \left[- \left(\frac{v}{c} \right)^k \right] \quad (12)$$

where, k is the shape parameter (dimensionless), c is the scale parameter in (m/s), and $f(v)$ is the Weibull probability distribution function. k and

Table 13

Component size and techno-economic metric of the configurations for ORIBA JUS.

ORIBA JUS					
Configuration	Grid/DG	Grid/DG/PV	Grid/DG/PV/Battery	Grid/DG/PV/battery/wind	Grid/DG/PV/wind
Grid (kW)	999,999	999,999	999,999	999,999	999,999
DG (kW)	320	320	320	320	320
PV (kW)		372	388	354	365
Converter (kW)		104	111	108	108
Battery (unit)			24	24	
Wind turbine (unit)				1	1
NPC (\$)	2,436,907	1,831,510.0	1,868,859.0	1,869,129.00	1,831,094
LCOE (\$/kWh)	0.109	0.082	0.084	0.084	0.082
CO ₂ emissions (kg/year)	467,539.0	317,767.00	313,349.00	313,918.00	314,663.0
Fuel (l/year)	6755.00	5124.00	4281.00	4325.00	5061.00
IRR (%)		12.40	10.50	10.70	11.90
SPBP (year)		7.51	8.76	8.63	7.75
RF (%)	0	31.9	32.80	32.70	32.60

Table 14

Component size and techno-economic metrics for AGHAZARMADRAN.

AGHAZARMADRAN					
Configuration	Grid/DG	Grid/DG/PV	Grid/DG/PV/battery	Grid/DG/PV/battery/wind	Grid/DG/PV/wind
Grid (kW)	999,999	999,999	999,999	999,999	999,999
DG (kW)	36	36	36	36	36
PV (kW)		62.5	64.7	24.9	23.7
Converter (kW)		25	28.9	11	10.7
Battery (unit)			24	24	
Wind turbine (unit)				5	5
NPC (\$)	458,153.60	262,446.60	312,801.80	259,244.90	206,927.6
LCOE (\$/kWh)	0.121	0.0692	0.082	0.068	0.055
CO ₂ emissions (kg/year)	79,780.00	42,831.00	41,499.00	18,271.00	19,178.00
Fuel (l/year)	1368.00	862.00	460.00	229.00	526.00
IRR (%)		21.10	11.20	9.50	14.20
SPBP (year)		4.66	6.94	9.82	6.76
RF (%)	0	46.3	48	80.7	79.9

c can be determined using the mean wind speed-standard deviation method. This is also known as the empirical method [28], which is related to the average wind speed as expressed in Eqs. (13) and (14) [28,29,32]:

$$c = \frac{v_{\text{mean}}}{\Gamma\left(1 + \frac{1}{k}\right)} \quad (13)$$

$$k = \left(\frac{\delta}{v_{\text{mean}}}\right)^{-1.086} \quad (1 \leq k \leq 10) \quad (14)$$

where Γ is the gamma function, δ is the standard deviation, and v_{mean} is the mean wind speed (m/s) over a period. In this work, the Bergey Excel 10 wind turbine has been selected as the wind turbine. Fig. 14 shows the power curve of the selected wind turbine and Table 9 outlines its features and characteristics.

2.4.6. Converter

Converters are electronic devices that adapt electrical power from DC to AC (inversion) and/or from AC to DC (rectification). The size of the converter is determined by the maximum amount of AC power that can be produced by inverting DC power (known as the inverter capacity). The user can specify the rectifier capacity as a percentage of the inverter capacity. Additionally, users must indicate if the inverter can work in parallel with an AC power source, such as a generator or grid. This synchronization requires the converter to match the frequency, maximum voltage, and phase sequences. In HOMER, the inverter properties include physical characteristics, such as inversion and rectification efficiencies, and economic properties, such as cost, annual O&M cost, and expected lifetime. The efficiency of the inverter and rectifier is

considered to be 95 %, and the rectifier's relative capacity is assumed to be 100 % of the inverter capacity.

2.4.7. Battery

Batteries are used to store excess electricity for later utilization, such as during nighttime or when the power produced by the system is insufficient to meet the load. The battery model gives its state of charge (SOC) and voltage at any time. The state of charge represents the remaining energy at any time in a given time. The SOC is assumed to be 100 % or 1.0 when the battery is fully charged and 0 % or 0.0 when the battery is fully discharged. The evolution of SOC over an interval of time $[t_0, t]$ can be determined as expressed as [33]:

$$z(t) = z(t_0) - \frac{1}{Q} \int_{t_0}^t \eta(\tau) * i(\tau) d\tau \quad (15)$$

The discrete-time form of Eq. (15) can be written as expressed in [33]:

$$z[k+1] = z[k] - \frac{\Delta t}{Q} \eta[k] * i[k] \quad (16)$$

The terminal voltage of the battery can be expressed as in Eq. (17):

$$V[k] = \text{OCV}(z[k]) - i[k] * R_{\text{int}} \quad (17)$$

where OCV is the open circuit voltage of the battery, R_{int} is the equivalent internal series resistance, and $v[k]$ is the voltage. The main characteristics of the battery are battery capacity, voltage, energy density/specific energy, self-discharge, autonomy, and cycling capacity.

2.4.7.1. Battery sizing. The sizing of the battery bank in a hybrid power system consists of determining the required number of batteries,

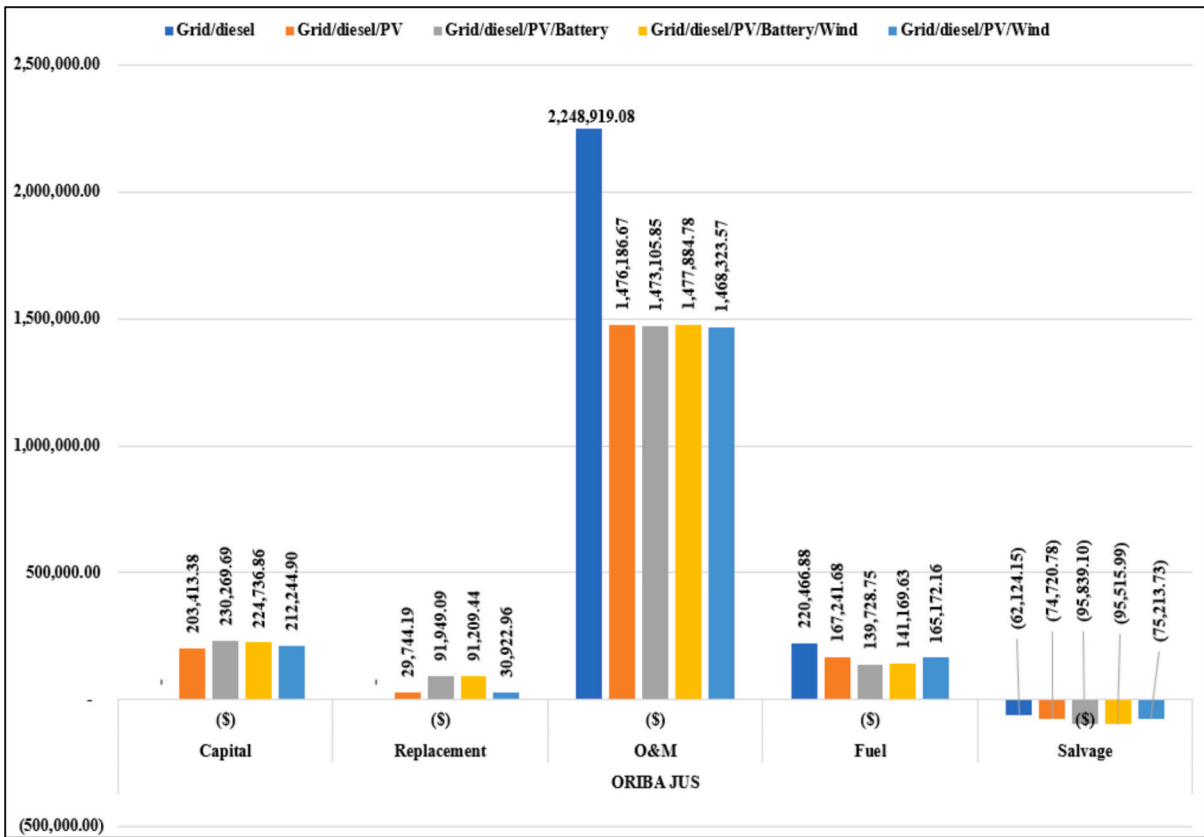


Fig. 19. Cost summary of the four different configurations for ORIBA JUS.

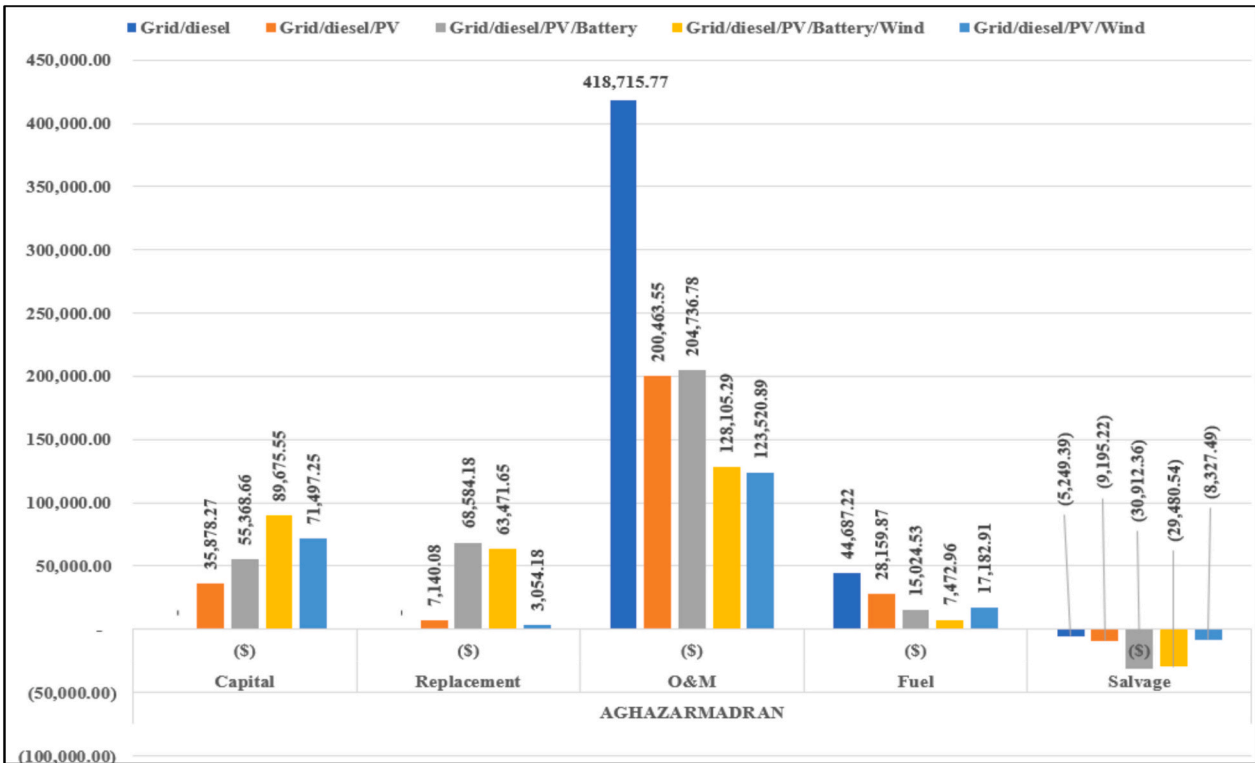


Fig. 20. Cost summary of the four different configurations for AGHAZARMADRAN.

Table 15

Average monthly power output for grid/diesel system for the two regions.

	Niamey		Agadez	
	DG (kWh/year)	Grid (kWh/year)	DG (kWh/year)	Grid (kWh/year)
Annual average	22,139	711,566	3907	120,566

configuration, and the individual capacity of the battery. Many factors affect the sizing of the battery bank such as the number of days of autonomy, maximum DoD (depth of discharge), the daily energy demand, battery round-trip efficiency, and the system voltage. The capacity of the battery bank is given in Eq. (18) as expressed in [21]:

$$C(\text{Ah}) = \frac{E_{\text{demand}} * N_{\text{aut}}}{\text{DoD} * U_{\text{sys}} * \eta_{\text{bat}} * \eta_{\text{inv}}} \quad (18)$$

where $C(\text{Ah})$ is the battery capacity (Ah), E_{demand} is the daily energy demand (Wh), N_{aut} is the number of days of autonomy, DoD the depth of discharge, η_{bat} , η_{inv} are the battery and inverter efficiencies, respectively, and U_{sys} is the system DC bus voltage. The number of strings and the number of batteries in one string are given in [21]:

$$N_{\text{string}} = \frac{C(\text{Ah})}{C_{\text{battery}}} \quad (19)$$

$$N_{\text{series}} = \frac{U_{\text{sys}}}{U_{\text{bat}}} \quad (20)$$

Then the total number of batteries is given by:

$$N_{\text{bat}} = N_{\text{string}} * N_{\text{series}} \quad (21)$$

In HOMER, a battery is modeled as a device that can store a fixed amount of DC electricity with limits on charging/discharging rate, depth of discharge, and cycling energy. The battery storage system has the potential to enhance the quality of the grid power while promoting the use of renewable resources. This study specifically chooses PowerSafe OPzV 2000 for this purpose. The characteristics of the battery are shown in Table 10.

2.5. Economic model

The economic viability of any energy source is determined by its ability to produce energy at a low cost. Net Present Cost and Cost of Energy are the two models used by HOMER to perform an economic analysis of a hybrid energy system [34]. The NPC, also known as the life-cycle cost, is a measure of the present value of all the costs involved in installing and operating a component over its lifetime minus the present value of all the revenues earned during that time. The NPC is given in Eq. (22) as:

$$C_{\text{NPC}} = \frac{C_{\text{ann,tot}}}{\text{CRF}(i, R_{\text{project}})} \quad (22)$$

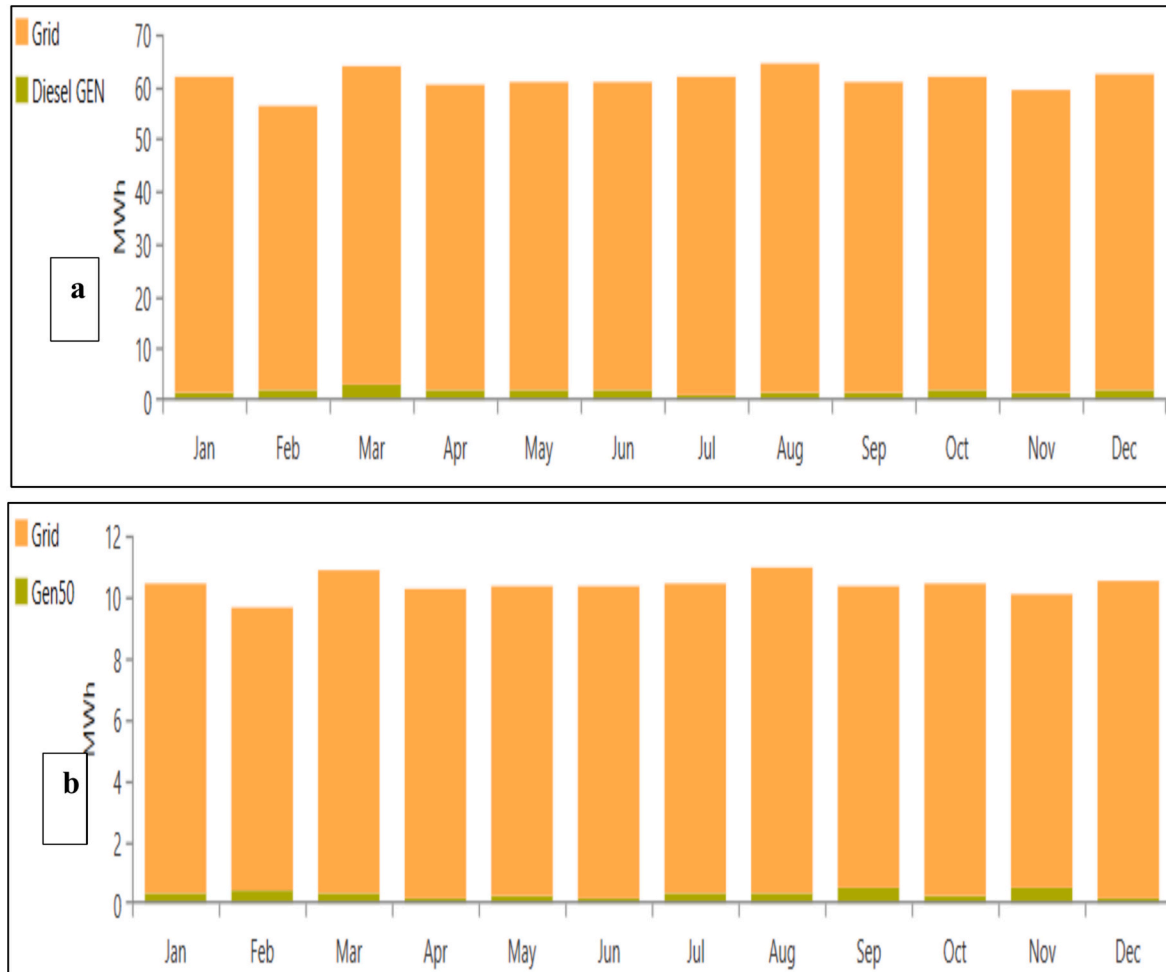


Fig. 21. Monthly electrical production for grid/diesel in (a) ORIBA JUS, (b) AGHAZARMADRAN.

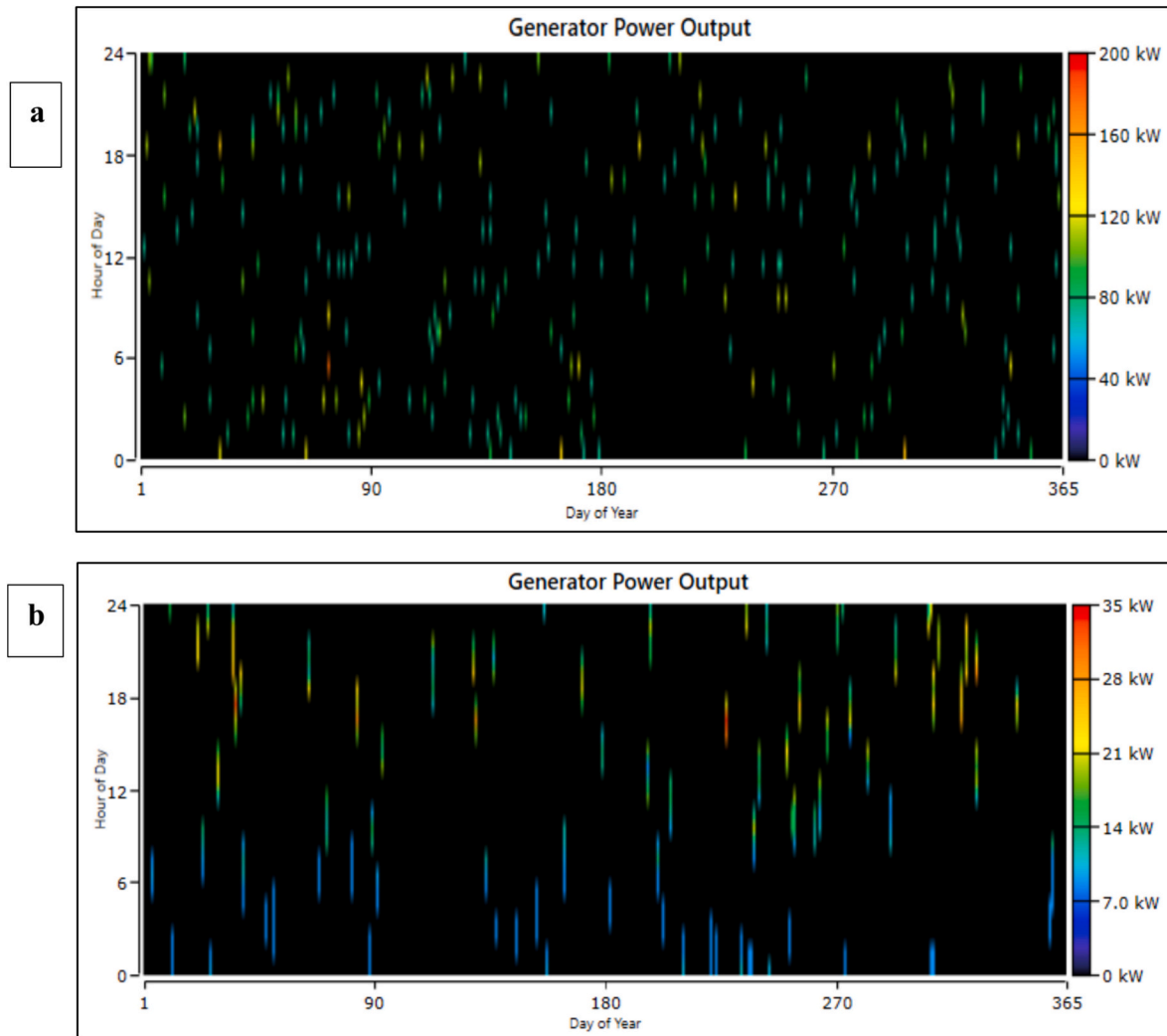


Fig. 22. Generator power output for (a) ORIBAS JUS, and (b) AGHAZARMADRAN.

Table 16

Average monthly power output for grid/diesel/PV in each company.

	ORIBAS JUS			AGHAZARMADRAN		
	PV (kW/year)	Grid (kW/year)	DG (kW/year)	PV (kW/year)	Grid (kW/year)	DG (kW/year)
Annual average	531,171	481,396	16,777	128,872	64,199	2393

where, $C_{ann,tot}$ is the total annualized cost, i is the annual real interest rate (the discount rate), $R_{project}$ is the lifetime of the project and the CRF is the capital recovery factor, which is defined in Eq. (23) as expressed in [34]:

$$CRF(i, N) = \frac{i(1+i)^N}{(1+i)^N - 1} \quad (23)$$

where, N is the number of years.

i , the real interest rate can be computed using Eq. (24):

$$i = \frac{i' - f}{1 + f} [\%] \quad (24)$$

where, f is the inflation rate [%], and i' is the nominal interest rate [%].

The Levelized Cost of Energy (LCOE) or COE is the average cost of producing one kilowatt-hour (kWh) of electricity by the system. It is defined by the Eq. (25) as expressed in [34]:

$$LCOE = \frac{C_{ann,tot}}{E_{prim} + E_{df} + E_{gri,sales}} \quad (25)$$

where, E_{pr} , E_{df} , $E_{gri,sales}$ are the total primary electrical load, total deferrable load that the system serves over one year, and total energy sold to the grid by the system over one year, respectively.

2.5.1. Cost structure of the components

A survey was conducted by the sellers and installers of PV system components in each region to build the cost structure of the components. The objective is to determine the initial cost per unit of each component. The replacement cost, operation and maintenance cost for each component are evaluated considering the assumptions in Table 11. Table 12 shows the cost of components and economic assumptions. Since wind turbines are not locally produced, the cost structure of wind turbines is calculated using data from the IRENA report [35–37].

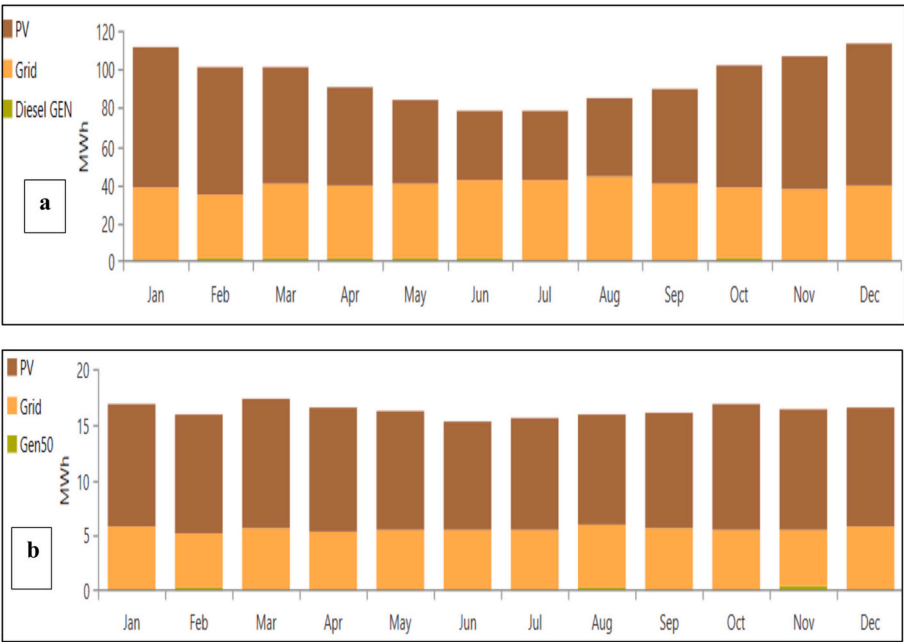


Fig. 23. Monthly electrical production for grid/diesel/PV for (a) ORIBA JUS, and (b) AGHAZARMADRAN.

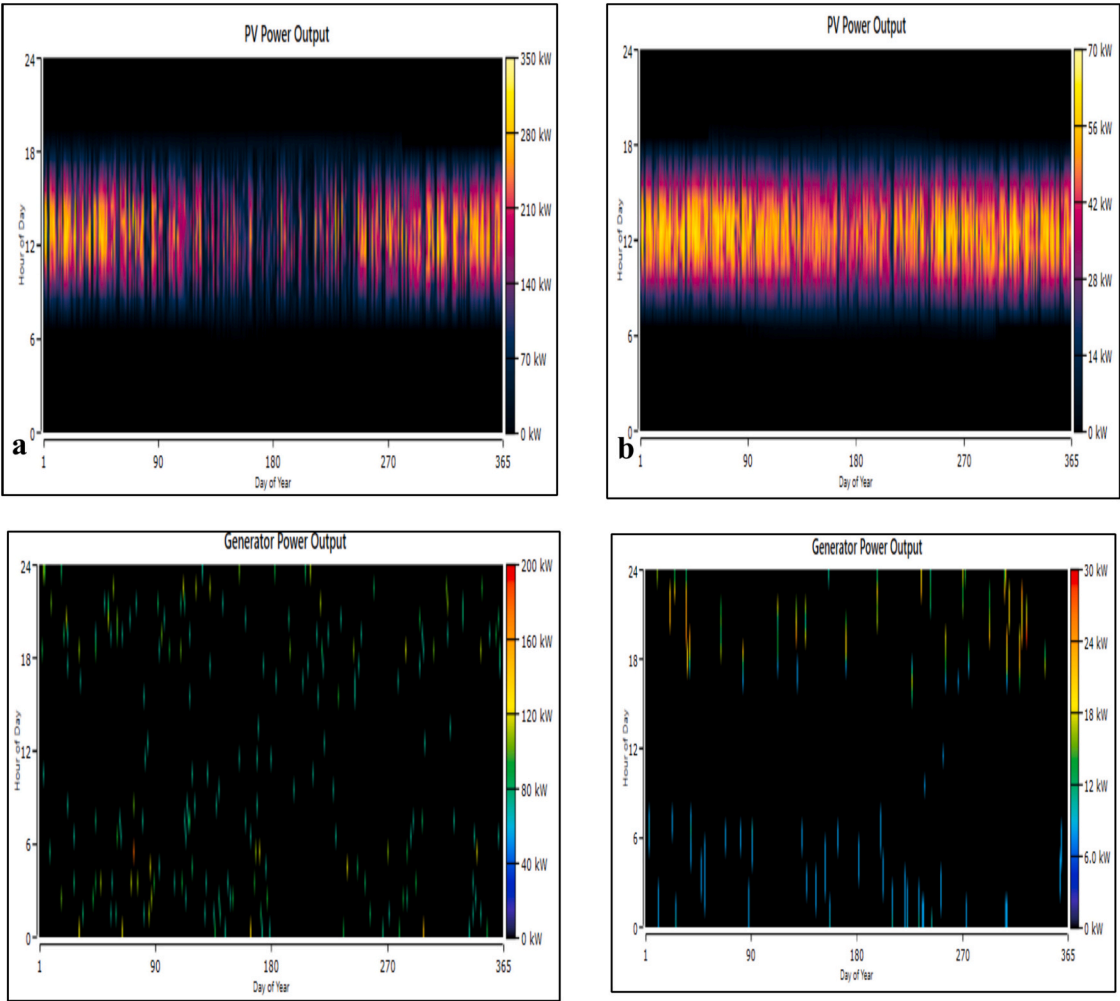
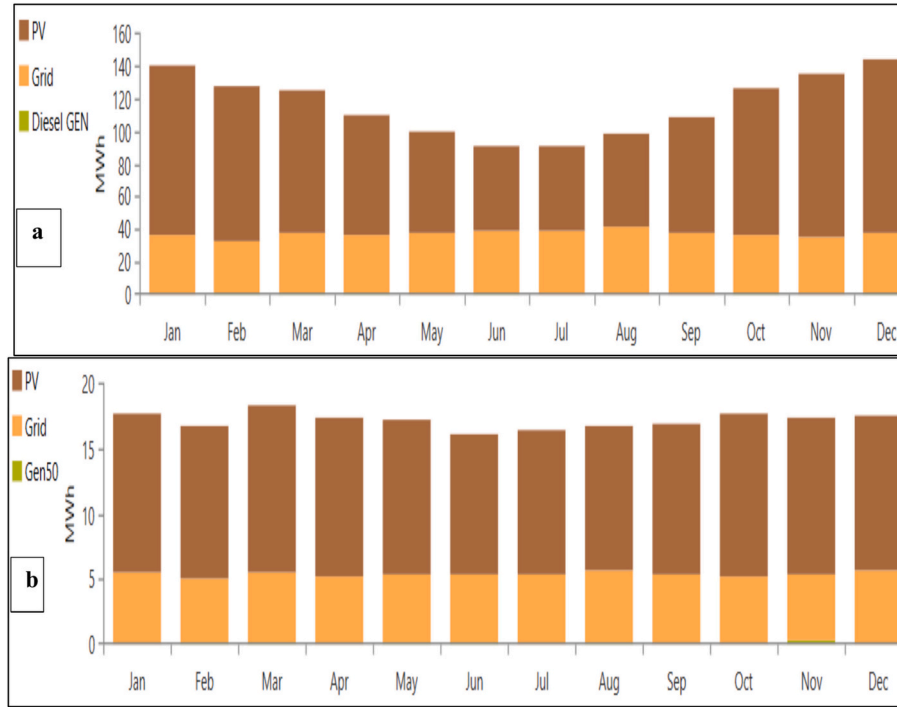


Fig. 24. Seasonal power output of diesel generator and PV array for (a) ORIBAS JUS, and (b) AGHAZARMADRAN.

Table 17

Average monthly power output for grid/diesel/PV/battery in each company.

	ORIBAS JUS				AGHAZARMADRAN			
	PV (kWh/year)	Grid (kWh/year)	DG (kWh/year)	Battery (kWh/year)	PV (kWh/year)	Grid (kWh/year)	DG (kWh/year)	Battery (kWh/year)
Annual average	554,242	477,925	13,719	1828	133,46	63,76	1281	1004

**Fig. 25.** Monthly electrical production with grid/diesel/PV/battery at (a) ORIBA JUS, (b) AGHAZARMADRAN.

2.6. Control strategy

HOMER determines the optimal HRES design to meet the load, based on the least cost system in terms of NPC. Several operational strategies are available in HOMER, such as cycle-charging (CC), load following (LF), combined dispatch strategy that combines CC and LF, and user-defined dispatch strategies which allow users to develop a control strategy using MATLAB. The LF and CC are the main control strategies in HOMER [38]. In the CC strategy, the diesel generator is utilized to supply the primary load while simultaneously charging the batteries. The charged batteries can later be employed to supply the load during periods of generator downtime or when the load demand exceeds the generator's capacity. In the LF control strategy, the diesel generator operates to match the power demand, and only renewable energy is used to charge the battery. In situations where the power demand is high and the battery energy storage (BES) needs to be charged, the diesel generator (DG) in the CC strategy runs at full power to fulfill the requirements. This is done by using surplus electrical production. On the other hand, the LF strategy is focused on maximizing renewable energy self-consumption. Its purpose is to ensure that renewable energy sources are used to their maximum potential. The CC strategy, however, is designed to prioritize reliability over renewable energy usage [17,38].

3. Results and discussions

3.1. Wind resource assessment

The annual average power density in Niamey at 10 m is 23.190 W/

m^2 . The minimum and maximum values for power density correspond to the minimum and maximum wind speeds observed in November and June, respectively, with values of 10.002 W/ m^2 and 38.325 W/ m^2 (Fig. 15a). On the other hand, the annual average power density for Agadez at 10 m is 60.241 W/ m^2 . The minimum and maximum values are 25.296 W/ m^2 and 104.978 W/ m^2 , respectively (Fig. 15b).

The monthly Weibull probability density functions for each region are shown in Fig. 16. It indicates the frequency of occurrence of wind speed for each month. In Niamey, the highest frequency of 0.498, occurs during October and corresponds to a wind velocity of 3 m/s, while in Agadez, it occurs in July and corresponds to a wind velocity of 4 m/s with a frequency of 0.373.

3.2. Simulations results

After all the data is inputted into HOMER, the simulation for each company is carried out. At the end of the simulation, HOMER creates a list of all the possible configurations and classifies them by category, from the least cost to the highest cost configuration. To select the optimal configuration for each region, the most cost-effective system will be compared with five different configurations based on the following metrics: grid/diesel is the base case because it is the pre-existing system, grid/diesel/PV, grid/diesel/PV/battery, grid/diesel/PV/battery/wind, and grid/diesel/PV/wind.

3.3. Results of simulation and optimization

This section presents the simulation results for each configuration.

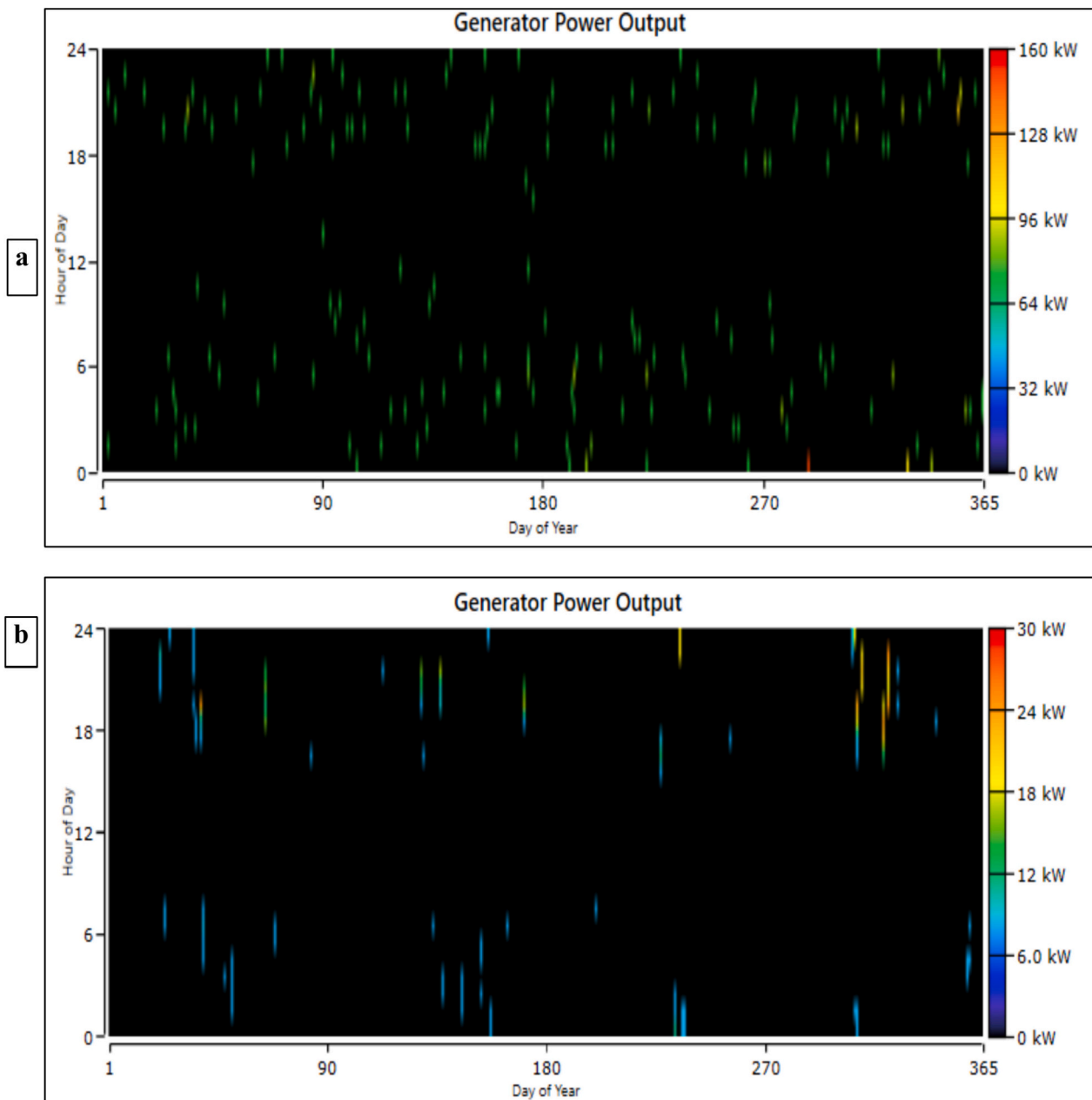


Fig. 26. Seasonal power output of DGs for grid/diesel/PV/battery at (a) ORIBAS JUS, and (b) AGHAZARMADRAN.

Table 18

Average annual production for grid/diesel/PV/battery/wind for each company.

	ORIBAS JUS					AGHAZARMADRAN				
	PV (kWh/year)	Grid (kWh/year)	DG (kWh/year)	Battery (kWh/year)	Wind (kWh/year)	PV (kWh/year)	Grid (kWh/year)	DG (kWh/year)	Battery (kWh/year)	Wind (kWh/year)
Annual average	505,562	478,642	13,853	1813	9376	51,425	27,962	590	475	110,644

For each company, the results of the system are all designed to supply the same load profile. HOMER can create the daily and seasonal variation of the load from a one-day sample load profile and by providing a day-to-day and timestep random variability. For this study, the 10 % and 20 % are considered for day-to-day and timestep random variability. Figs. 17 and 18 show the seasonal and yearly critical load profiles, respectively, for each company. Tables 13 and 14 show the techno-economic metrics and component size of all the configurations

considered in this study for ORIBA JUS and AGHAZARMADRAN. Grid/diesel/PV/wind was found to be the least cost system for ORIBA JUS and AGHAZARMADRAN. The cost summary of the different configurations for ORIBA JUS and AGHAZARMADRAN is depicted in Figs. 19 and 20, respectively. For the base case, the CC equals zero as it is the pre-existing system. The figures show that the O&M cost is the highest expenditure of all the systems in the two companies.

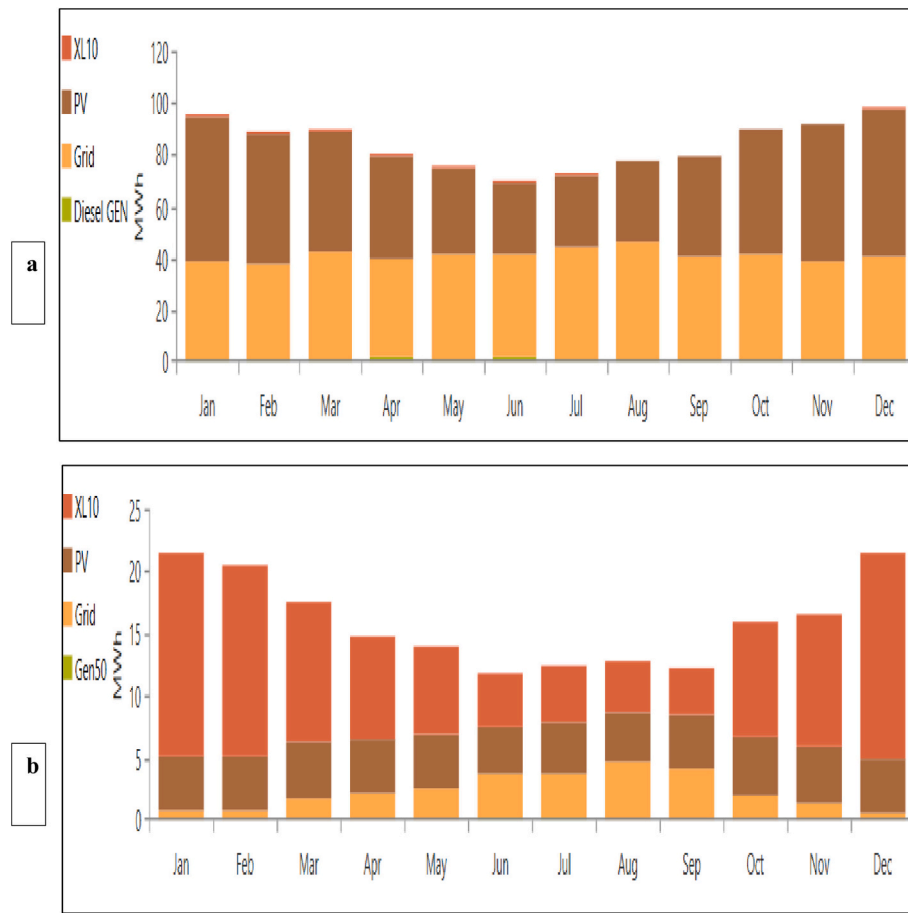


Fig. 27. Monthly electrical production per components at (a) ORIBA JUS, and (b) AGHAZARMADRAN.

3.4. Discussions

3.4.1. Configuration based on grid/diesel system

In this configuration, the grid is the main power source. The diesel generator is working as a backup to support critical and non-critical loads during grid failure. The total electricity production of the grid/diesel system for each company is presented in Table 15. Fig. 21 shows the monthly electrical production for each company. As indicated by HOMER from the simulation, the grid constitutes 96.9 % with a total electricity production of 711,566 kWh/year for ORIBA JUS and 120,566 kWh/year for AGHAZARMADRAN.

From the data shown in Tables 13 and 14, the grid/diesel system configuration has the highest NPC and LCOE among all the configurations for each company. The NPC and LCOE for ORIBAS JUS are \$2,436,907.00 and 0.109 \$/kWh, respectively, while they are \$458,153.60 and 0.121 \$/kWh, respectively, for AGHAZARMADRAN. The yearly fuel consumption and the CO₂ emissions are 6755.00 l and 467,539 kg respectively, for ORIBA JUS, and 1368.00 l and 79,780 kg for AGHAZARMADRAN. In line with Figs. 19 and 20, this system also has the highest O&M cost, which accounts for \$32,248,919.08 for ORIBA JUS and \$418,715.77 for AGHAZARMADRAN. Fig. 22 depicts the seasonal power output of the generator in each company for the base case. For ORIBAS JUS, the working frequency per year of the generator is important, but the working time per day is short, while for the generator in AGHAZARMADRAN, the working time per day is long, but the starting frequency is not important. This is related to the grid outage data presented.

3.4.2. Configuration based on grid/diesel/PV system

In this configuration, the PV is used to supply the critical load during

the daytime. When there is an outage or when the PV output is not enough, and the grid is not available, the generator supplies the critical load. The component size of the optical system is shown in Tables 13 and 14. The PV array and converter with respective capacities of 372 kW and 104 kW for ORIBA JUS, and 62.5 kW and 25 kW for AGHAZARMADRAN are added to the base case. The techno-economic and environmental metrics for ORIBA JUS are shown in Table 13, and AGHAZARMADRAN in Table 14. The NPC and LCOE for ORIBA JUS are \$ 1,831,510.00 and 0.082 \$/kWh, respectively, and \$ 262,446.60 and 0.069 \$/kWh for AGHAZARMADRAN. The yearly fuel consumption and the CO₂ emissions are respectively equal to 5124 l and 317,767.00 kg for ORIBA JUS, and 862.00 l, and 42,831 kg for AGHAZARMADRAN. In line with Figs. 19 and 20, the grid/diesel/PV presents the lowest capital cost among all the configurations considered in this study for the two companies, which account for \$203,413.38 for ORIBA JUS and \$35,878.27 for AGHAZARMADRAN. The total electricity production per year for each component is shown in Table 16. For the generator, there is a decrease in the working periods during the daytime as the PV substitutes the generator. Fig. 23 represents the monthly electricity production per component. The monthly production of the diesel generator is almost insignificant. The power output of the PV and generator is shown in Fig. 24. The PV shows low production in Niamey, while production is almost constant in Agadez. For the generator, there is a decrease in the working periods during the daytime as the PV substitutes the generator.

3.4.3. Configuration based on grid/diesel/PV/battery system

Among all the configurations considered, the grid/diesel/PV/battery system is the most adopted and the most developed in the country. Table 17 shows the total electricity production per component for the two companies. Compared to the grid/diesel/PV, the annual electricity

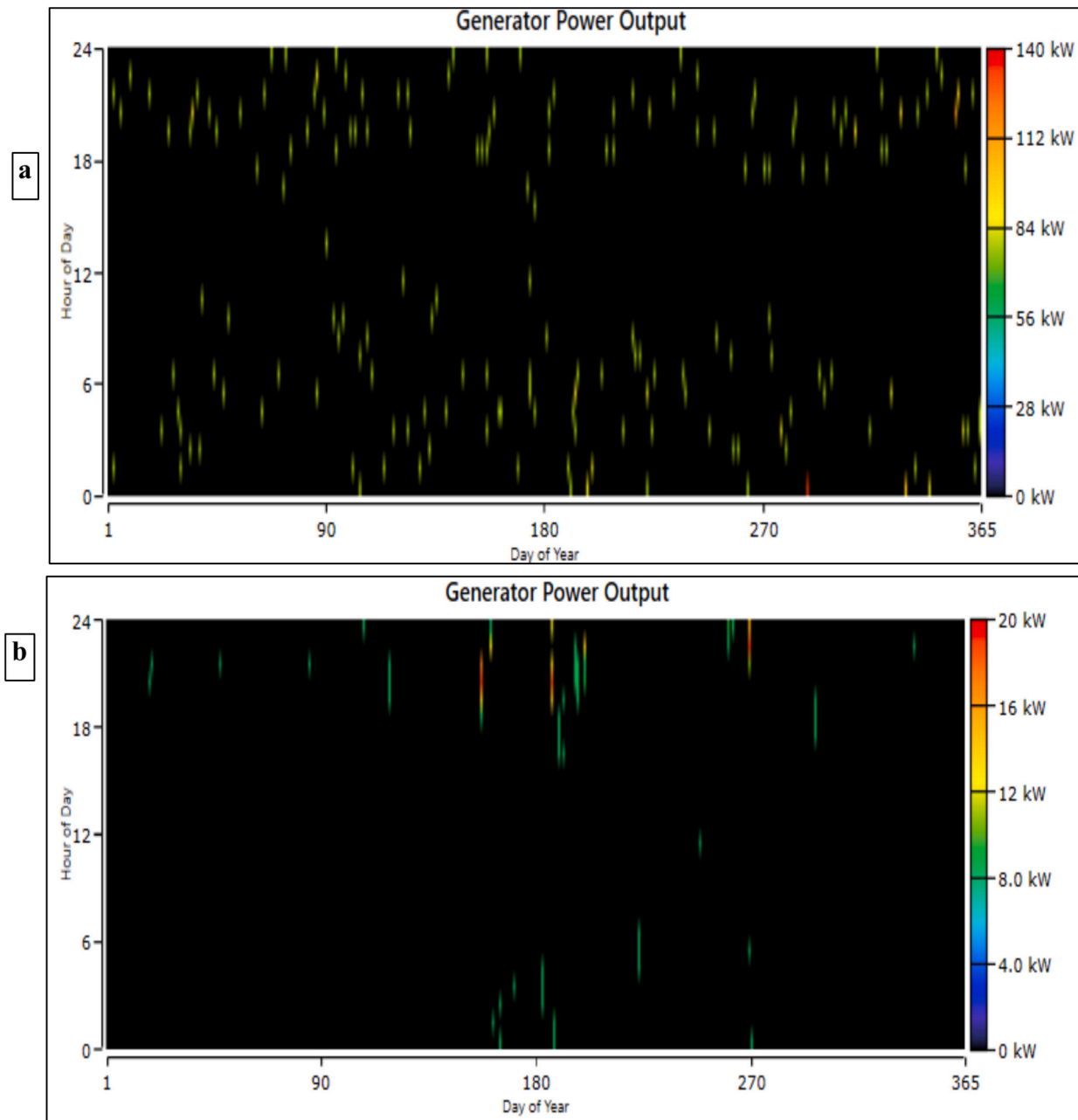


Fig. 28. Seasonal power output of DGs for grid/diesel/PV/battery/wind at (a) ORIBAS JUS, and (b) AGHAZARMADRAN.

production of PV for both companies increased from 531,171 kWh/year to 554,242.00 kWh/year for ORIBAS JUS and from 128,872 kWh/year to 133,463 kWh/year for AGHAZARMADRAN due to the incorporation of battery, while the annual mean grid purchase decreased slightly for the two companies. A significant decrease of around 18 % and 46 % in the annual production of the diesel generator is observed for ORIBAS JUS and AGHAZARMADRAN, respectively, when compared to grid/diesel/PV without battery. Fig. 25 shows the monthly electricity produced for each company.

It can be observed that the cost summary for the two companies from Figs. 19 and 20 shows that the O&M cost represents the main expenditure for both companies. The techno-economic and environmental parameters and component size for ORIBAS JUS and AGHAZARMADRAN are shown in Tables 13 and 14, respectively. The NPC and LCOE, as simulated by HOMER, are \$1,868,859.00 and 0.084 \$/kWh, respectively, for ORIBAS JUS, and \$312,801.80 and 0.082 \$/kWh for AGHAZARMADRAN. Fig. 26 represents the seasonal output power of the

generator in the two companies for the grid/diesel/PV/battery configuration. Fig. 26 shows how the working frequency of generators is consequently reduced and shifted during the night across the year, especially in the case of AGHAZARMADRAN.

3.4.4. Configuration based on grid/diesel/PV/battery/wind system

The annual electricity production per component is shown in Table 18. Among all the configurations, the grid/diesel/PV/battery/wind system presents the lowest production of grid and diesel generators for AGHAZARMADRAN due to the high electricity production cost from wind turbines, representing 58 % of the total electricity production of the system. For ORIBAS JUS, the grid purchase and PV are the main power sources, while the shares of generator and wind turbine are insignificant. When compared to the grid/diesel/PV/battery system, the annual production from batteries decreased in both companies.

The component sizes and techno-economic parameters for this configuration at the two companies reveal that the NPC and LCOE values

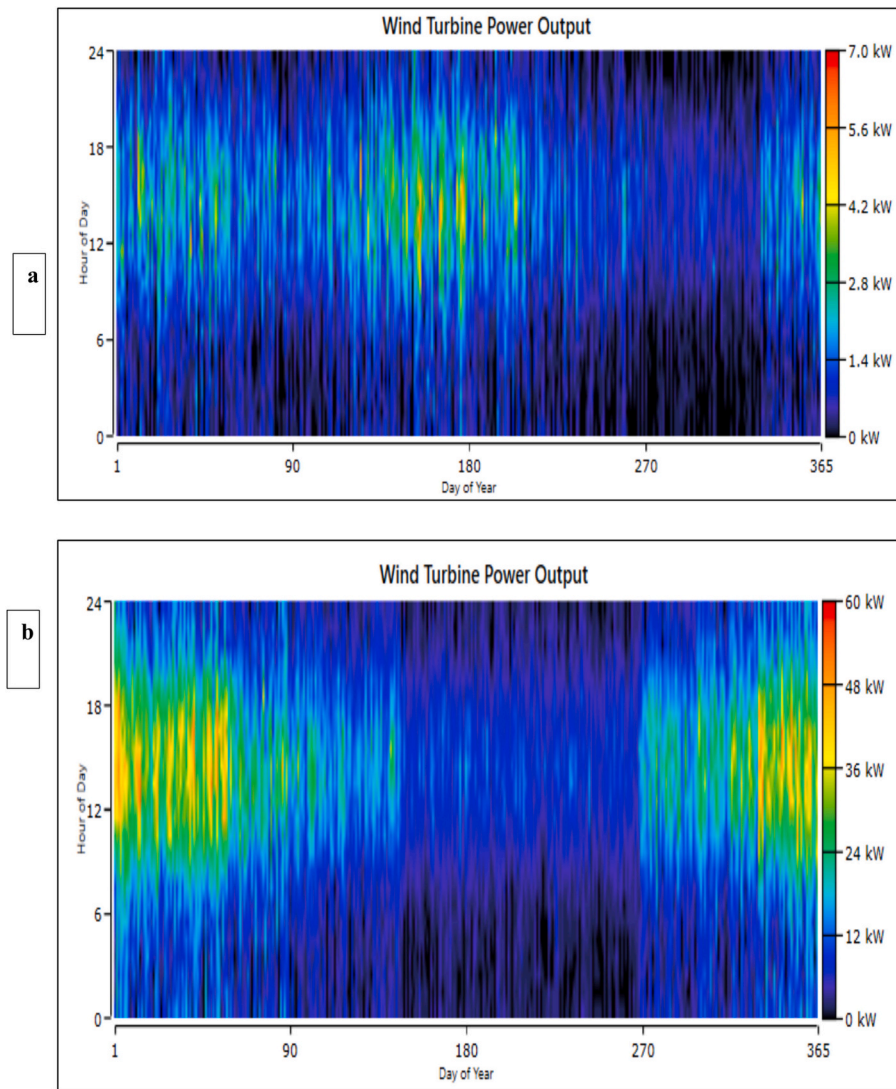


Fig. 29. Seasonal power output of wind turbines for grid/diesel/PV/battery/wind at (a) ORIBAS JUS, and (b) AGHAZARMADRAN.

are \$ 1,869,129.00 and 0.084 \$/kWh, respectively, for ORIBA JUS, and \$ 259,244.90 and 0.068 \$/kWh for AGHAZARMADRAN (Tables 13 and 14). From Figs. 19 and 20, it can further be seen that this configuration presents the highest capital cost in the case of AGHAZARMADRAN due to the inclusion of wind turbines, but the O&M cost is largely lower than the previous configuration due to reduction in the generator working hours, as shown in Figs. 27 and 28. For ORIBA JUS, the grid/diesel/PV/battery/wind system has almost a similar cost distribution as the previous grid/diesel/PV/battery system. The capital cost is \$224,736.86 for ORIBA JUS and \$ 89,675.55 for AGHAZARMADRAN. The output of the wind turbine shown in Fig. 29 reveals that in Niamey, the wind potential is very low; most of the time, especially during nighttime, the wind production is between 0 and 1.4 kW. The production is also very low in September, October, and November. For Agadez, wind turbine production is very important during the daytime, but with a reduction between June and September.

3.4.5. Configuration based on grid/diesel/PV/wind system

In this configuration, PV and wind turbines are used to supply the critical loads in case of grid outages and are supported by the diesel generator when their output power is not enough. Wind turbine represents the main power source for AGHAZARMADRAN (Fig. 30). It accounts for 58.5 % of the total electricity production, followed by PV,

which accounts for 25.9 %. For ORIBA JUS, the electricity production distribution remains similar to the grid/diesel/PV/battery/wind. The total electricity production per component is presented in Table 19. The output power of the generator is depicted in Fig. 31. This system is the most cost-effective configuration for both companies regarding the NPC and LCOE, as shown in Tables 13 and 14.

3.5. Sensitivity analysis

The diesel price, solar radiation, wind speed, and an increase in load demand are considered for the sensitivity analysis in this study. The configurations based on grid/diesel/PV/battery/wind, and grid/diesel/PV/wind are considered for the sensitivity analysis for ORIBA JUS and AGHAZARMADRAN, respectively.

3.5.1. The effect of diesel price

The diesel price varies based on the country's historical price. It varies from \$1 to \$1.3 per litre. The results of the sensitivity analysis based on HOMER show that the optimal system configuration does not change. The result in Fig. 32 indicates that the NPC and LCOE increased with an increase in diesel price for both ORIBA JUS and AGHAZARMADRAN.

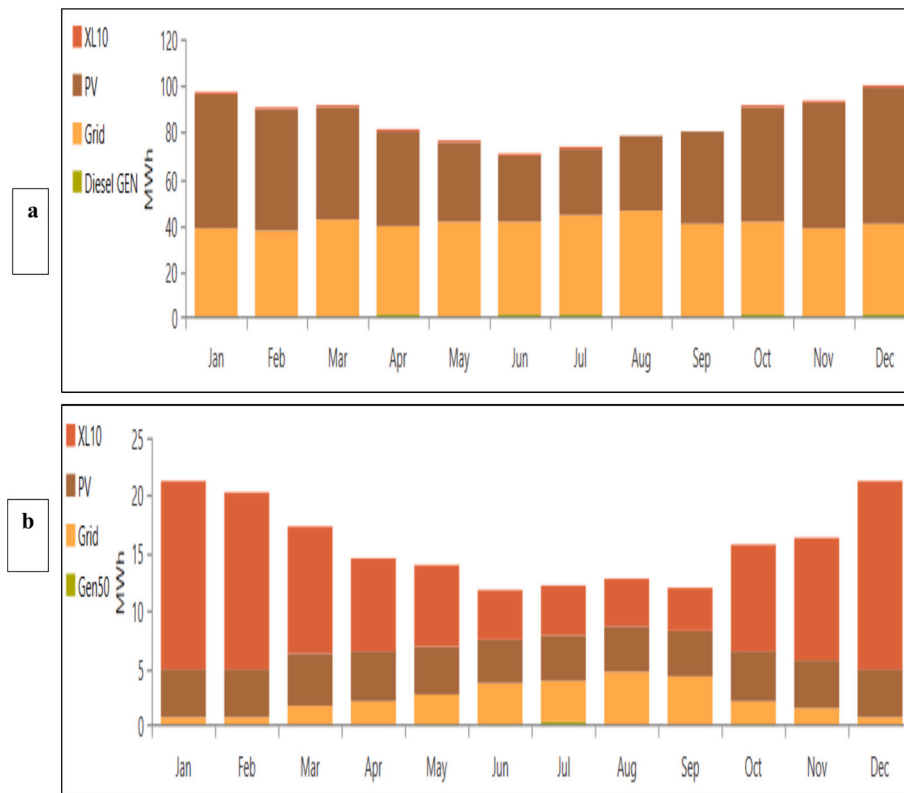


Fig. 30. Monthly electrical production per components at (a) ORIBA JUS, and (b) AGHAZARMADRAN.

Table 19

Average annual production for grid/diesel/PV/wind of each company.

	ORIBAS JUS				AGHAZARMADRAN			
	PV (kWh/year)	Grid (kWh/year)	DG (kWh/year)	Wind (kWh/year)	PV (kWh/year)	Grid (kWh/year)	DG (kWh/year)	Wind (kWh/year)
Annual average	520,431	476,749	16,560	9376	48,932	28,166	1366	110,644

3.5.2. The effect of durations of grid outages

The duration of grid outage varies from 120 to 180 h/year for ORIBA JUS and from 240 to 280 h/year for AGHAZARMADRAN, while other variables are kept constant. In this sensitivity analysis, the frequency of grid outages varies proportionally with the outage duration, and only unscheduled outage was considered. Fig. 33 shows the effect of duration of grid outage on the NPC and LCOE of the optimal system. For ORIBA JUS, the NPC and LCOE remain constant with an increase in the interruption duration, while for AGHAZARMADRAN, NPC and LCOE increase when the average grid interruption duration increases.

3.5.3. The effect of solar radiation

Fig. 34 shows the effect of change in solar radiation on the NPC and LCOE of the optimal system in both companies. Solar radiation is varied from 4 to 5.5 kWh/m²/day for ORIBA JUS and from 6 to 7.5 kWh/m²/day for AGHAZARMADRAN. There is a reduction in NPC and LCOE for ORIBA JUS and AGHAZARMADRAN when the daily solar radiation increases. This justifies the weather dependency of hybrid renewable energy systems.

3.5.4. The effect of wind speed

In the two companies, the NPC and LCOE decrease with reduction in the average wind speed as shown in Fig. 35. This shows that a cost-effective hybrid system with wind turbine is very sensible to changes in wind speed.

3.5.5. The effect of load variation

The critical loads considered for ORIBA JUS and AGHAZARMADRAN were 2004.4 kWh/day and 340.65 kWh/day, respectively; due to the variability associated with load, a sensitivity analysis is necessary to assess the effect of load on the optimal system performances. The load for ORIBA JUS varies between 1800 and 2400 kWh/day, while it varies between 320 and 400 kWh/day for AGHAZARMADRAN. Fig. 36a shows that the LCOE decreases for ORIBA JUS with the increase in the load, but for AGHAZARMADRAN (Fig. 36b), a slight increase of the LCOE between 360 and 380 kWh/day was observed; the NPC was found to increase with load variation in both companies. This sensitivity analysis reveals that the integration of RE is more beneficial in terms of LCOE for companies with high daily loads.

In addition to the sensitivity analysis, the impact of renewable energy potential on the optimal system configuration and LCOE is investigated in both companies. In the Niamey region (Fig. 37a), the grid/diesel/PV appeared to be the optimal system design when the wind speed drops below 3.3 m/s at 10 m. This reveals the weakness of the wind potential in the Niamey region, as the average monthly wind speed varies between 2.54 and 3.97 m/s. When the solar radiation increases from 4 to 5.5 kWh/m²/day, the LCOE of grid/diesel/PV varies slightly from 0.083 to 0.08 \$/kWh. Fig. 37b indicates that grid/diesel/PV/wind remains the optimal configuration for the Agadez region when the wind speed varies from 4 to 5.5 m/s at 10 m and solar radiation from 6.3 to 7.5 kWh/m²/day. The LCOE can further decrease to 0.39 \$/kWh with the maximum values of wind speed and solar radiation.

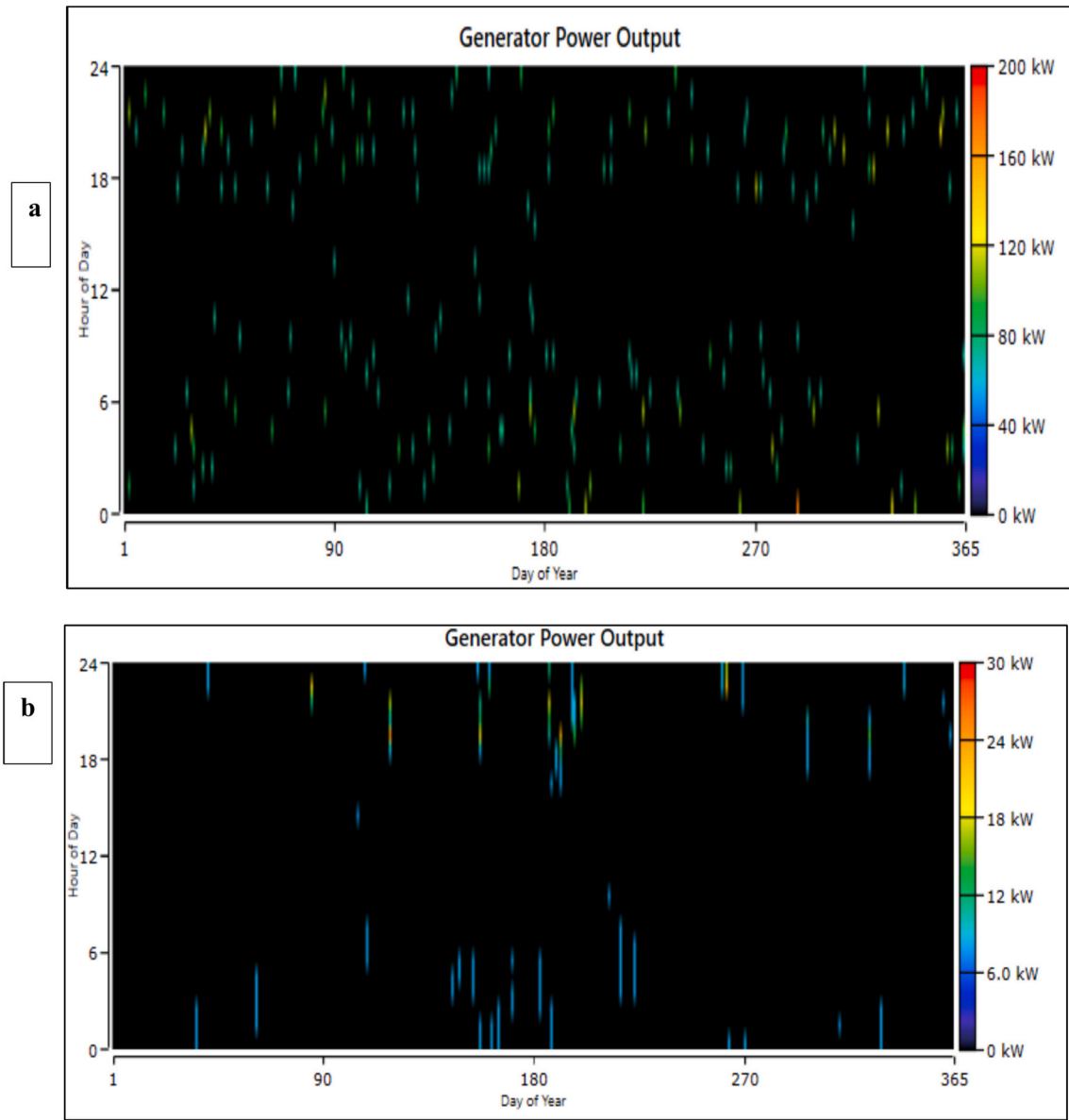


Fig. 31. Output power of wind turbine and diesel generator of grid/diesel/PV/wind hybrid system for (a) ORIBA JUS, and (b) AGHAZARMADRAN.

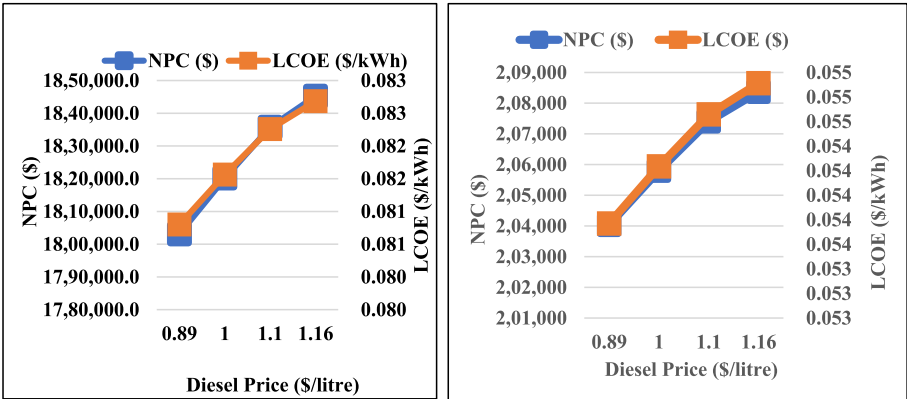


Fig. 32. Effect of diesel price on NPC and LCOE at (a) ORIBA JUS, and (b) AGHAZARMADRAN.

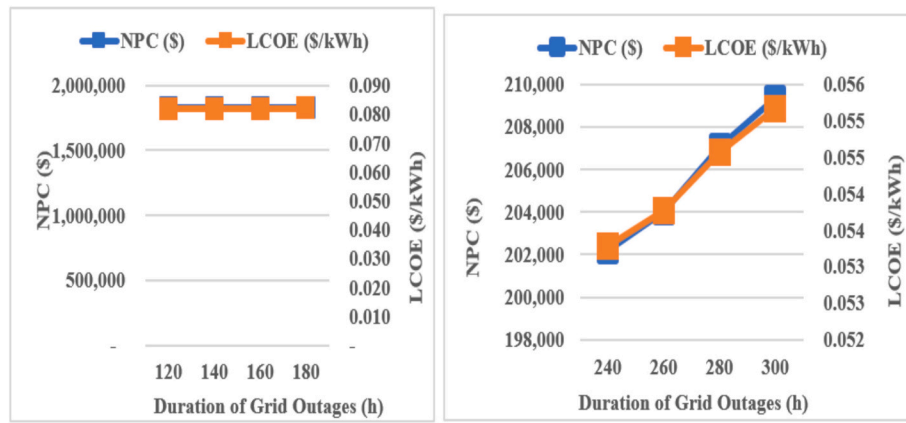


Fig. 33. Effect of grid outages duration on NPC and LCOE at (a) ORIBA JUS, and (b) AGHAZARMADRAN.

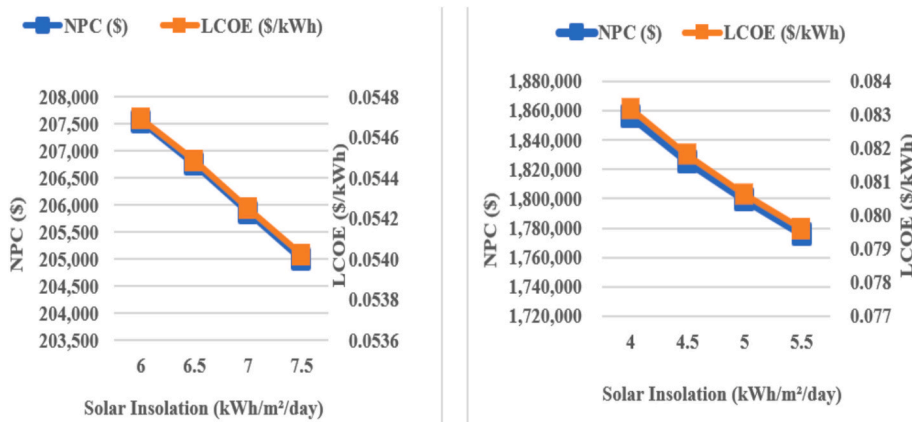


Fig. 34. Effect of solar radiation on NPC and LCOE at (a) ORIBA JUS, and (b) AGHAZARMADRAN.

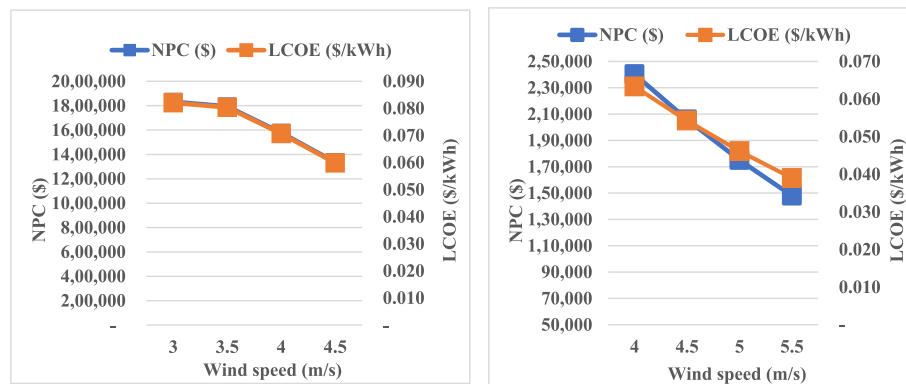


Fig. 35. Effect of wind speed on NPC and LCOE at (a) ORIBA JUS, and (b) AGHAZARMADRAN.

3.6. Comparison of the techno-economic and environmental assessments of the proposed systems

3.6.1. Techno-economic assessment

All the configurations presented in this study can meet all the critical loads of each company, cost-effectively. One of the objectives of this study is to identify the most cost-effective system for each region. To achieve this objective, this section evaluates the configurations discussed in Section 3.4, and compares them from an economic and environmental perspective. The hybrid grid/diesel/PV/wind system is the most cost-effective of all the different configurations considered for the

two companies in terms of NPC and LCOE. For ORIBAS JUS, the techno-economic metrics in Fig. 38 indicate that all the proposed systems gave an NPC (between 23 and 25 %) lower than the base case. The grid/diesel/PV/wind system has the lowest NPC of \$1,831,094.00 among all the configurations. However, this value is slightly different from the NPC of grid/diesel/PV, which is equal to \$1,831,510.00. On the other hand, the LCOE of all the proposed systems varies between 0.082 and 0.084 \$/kWh, which represents a reduction of \$0.027 to \$0.03 of the LCOE of the base case. The grid/diesel/PV and grid/diesel/PV/wind systems show the same values of LCOE of 0.082\$/kWh, which is \$0.011 and \$0.068 lower than the grid electricity tariff during the non-peak

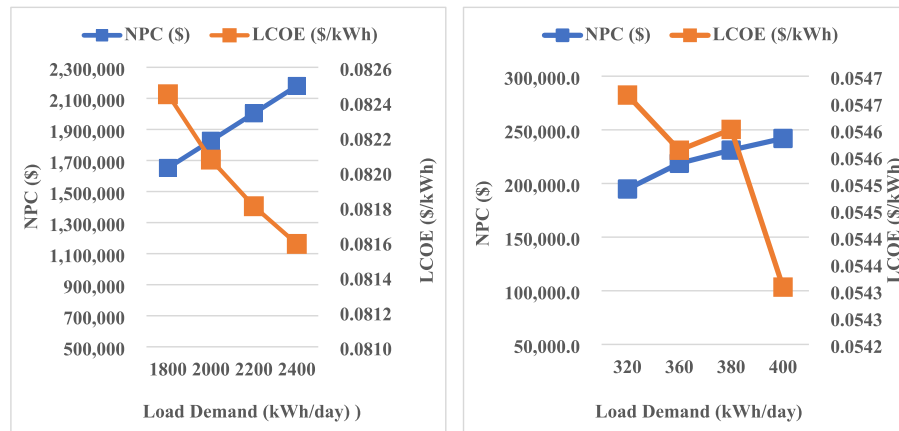


Fig. 36. Effect of load variation on NPC and LCOE at (a) ORIBA JUS, and (b) AGHAZARMADRAN.

demand period and peak demand period, respectively. In the same way, grid/diesel/PV/battery and grid/diesel/PV/battery/wind present similar values of LCOE. When it comes to the IRR and Simple Pay Back Period (SPBP), all the configurations gave an interesting value. However, the grid/diesel/PV appears to have the highest IRR and the lowest SPBP.

Based on Fig. 39, the configuration with the lowest cost for AGHAZARMADRAN is grid/diesel/PV/wind. It has a NPC of \$206,927.60 and a LCOE of 0.055 \$/kWh. The NPC of this system is 55 % lower than the NPC of the grid/diesel system, and also 21 %, 34 %, and 20 % lower than that of grid/diesel/PV, grid/diesel/PV/battery, and grid/diesel/PV/battery/wind, respectively, whereas the LCOE is about 0.07 \$ lower than the LCOE of grid/diesel which represent over 50 %. When compared to the grid price, the LCOE of grid/diesel/PV/wind is found to be \$0.095 and 0.04 lower than the rate during peak demand and non-peak periods, respectively, thus representing a reduction of 63 % and 41 %. The IRR and SPBP of the system are 14.2 % and 6.76 years, respectively; these are also competitive values because the values for IRR range from 9.5 % with grid/diesel/PV/battery/wind to 21.10 % with grid/diesel/PV, while the SPBP is from 4.66 years with grid/diesel/PV to 9.82 years with grid/diesel/PV/battery/wind.

3.6.2. Environmental assessment

The CO₂ emissions of the various systems for the two different locations were obtained from HOMER. For ORIBA JUS, the CO₂ emissions for each system as shown in Fig. 38, reveal that grid/diesel has the highest emissions of 467,539.00 kg/year due to the high fuel consumption of 6755.00 l/year, resulting from the long working hours of the generator. The grid/diesel/PV/battery has the lowest emissions of 313,349.00 kg/year, which represents a 33 %, 1.39 %, 0.18 %, and 0.4 % reduction of emissions when compared to the grid/diesel, grid/diesel/PV, grid/diesel/PV/battery/wind, and grid/diesel/PV/wind configurations, respectively. Consequently, this resulted in the reduction of fuel consumption of about 2474 l/year from the base case.

For AGHAZARMADRAN, the lowest CO₂ emission of 18,271.00 kg/year is observed with the grid/diesel/PV/battery/wind configuration (Fig. 39). This is about 907 kg/year lower than the grid/diesel/PV/wind configuration, which was found to be the optimal system based on the economic metrics. The CO₂ emissions for the grid/diesel/PV/wind configuration were about 76 % lower than the base case configuration, and 55 % and 54 % lower than the emissions observed with grid/diesel/PV and grid/diesel/PV/battery, respectively. This is explained by the important reduction in fuel consumption of around 842 l/year, when compared to the base case.

4. Conclusions and recommendations

This paper analyzed the feasibility of improving electricity reliability for selected industries in two different regions of Agadez and Niamey in Niger Republic through the integration of PV/wind hybrid systems, while taking into account unreliable grid/diesel systems. The study incorporated various metrics for evaluating system performance, including NPC, LCOE, IRR, SPBP, and CO₂ emissions. These metrics provide a comprehensive framework for assessing the economic feasibility and environmental impact of renewable energy systems. The NPC aids in understanding the total lifecycle costs, while LCOE allows for a comparison of costs across the different energy generation architectures.

Furthermore, the analysis of IRR and SPBP considers the financial viability of investments in renewable energy, which is crucial for industrial stakeholders looking for sustainable options. The evaluation of CO₂ emissions offers insight into the environmental benefits of shifting towards hybrid systems, highlighting the reduction in greenhouse gas emissions when compared to the traditional grid and diesel systems.

Based on the objectives and results of this research, the following conclusions were drawn:

- The solar potential is feasible in the two regions that were considered. The analysis of the meteorological data shows an annual mean solar radiation potential of 5.94 kWh/m²/day for Niamey, and 6.38 kWh/m²/day for Agadez. The mean annual maximum temperature in Niamey is 36.74 °C, while it is 36.07 °C in Agadez.
- The exploitation of wind is more feasible in Agadez than in Niamey. The wind potential assessment indicates an annual average speed of 3.304 m/s and 4.484 m/s at 10 m in Niamey and Agadez, respectively. The power density at 10 m is found to be 23.190 w/m² (Niamey) and 60.241 w/m² (Agadez).
- Among the configurations considered for the two selected companies, it is noticeable that the grid/diesel option will incur the highest costs. For ORIBA JUS in Niamey, the NPC amounts to \$2,436,907.00, and for AGHAZARMADRAN in Agadez region, NPC is \$458,153.60. In addition, this option also has the highest LCOE. When compared to the national grid price, it is apparent that for both regions, the LCOE exceeds the grid price of 0.093 \$/kWh for the non-peak hour period and lower than the peak load grid rate of 0.15 \$/kWh.
- The optimal system for ORIBA JUS (Niamey) was found to be the grid/diesel/PV/wind architecture, which included 365 kW of PV, 108 kW of the converter, and a wind turbine. The most suitable system for AGHAZARMADRAN in Agadez region is grid/diesel/PV/wind, which comprised 23.7 kW of PV, 10.7 kW of the converter, and five wind turbines.

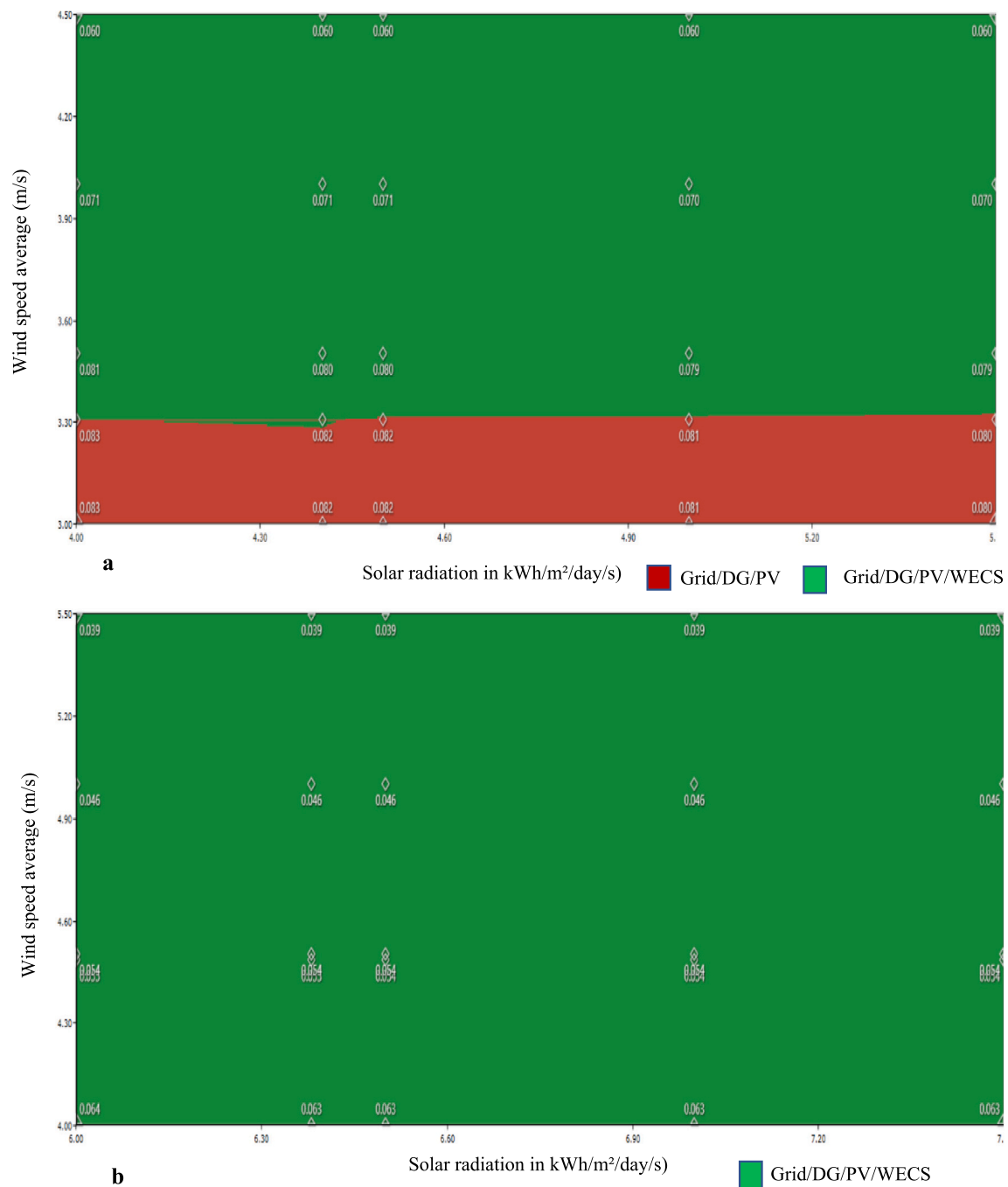


Fig. 37. Type of optimal systems and LCOE at (a) ORIBA JUS, and (b) AGHAZARMADRAN.

- v. Based on the sensitivity analysis, it was found that the NPC and LCOE of the optimal system, i.e., grid/diesel/PV/wind, remain competitive despite the variation in diesel price, grid outage duration, electric load, solar radiation, and wind speeds. However, the sensitivity analysis of the renewable energy potential suggests that for ORIBA JUS, the grid/diesel/PV/wind system can be replaced by grid/diesel/PV configuration, as depicted in Fig. 38. On the other hand, in Agadez regions, the grid/diesel/PV/wind system operates satisfactorily with varying wind speeds and solar radiation.

This study shows that the integration of solar PV and wind systems into the present grid and diesel systems can be both viable economically and environmentally in the North and South-East regions of Niger

Republic. The results of this study can serve as a guide for industrial owners, renewable energy developers, individuals, private organizations, and government bodies at various levels who are interested in improving electricity reliability through the integration of renewable energy sources in Niger.

Electricity reliability is a critical factor in the socio-economic progress of a nation. It is not only essential for new business development but also for the survival of existing firms in the industry [2]. Countries with abundant RE resources can overcome the challenges of power unreliability by integrating HRES into their energy infrastructure. To promote the integration of RE in developing countries like Niger, the following recommendations can be helpful:

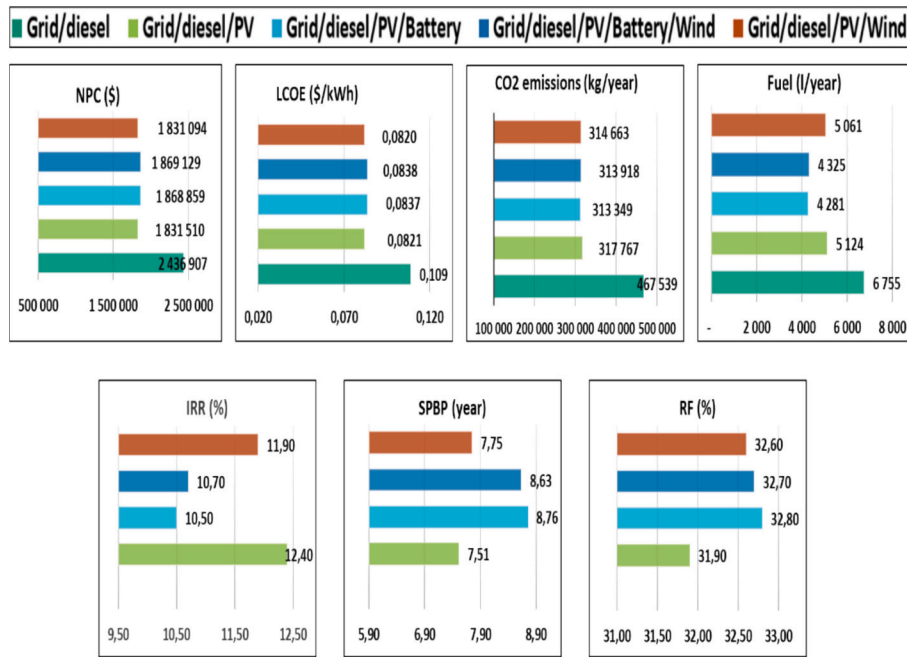


Fig. 38. Comparison of the techno-economic and environmental assessment of the different configurations for ORIBA JUS.

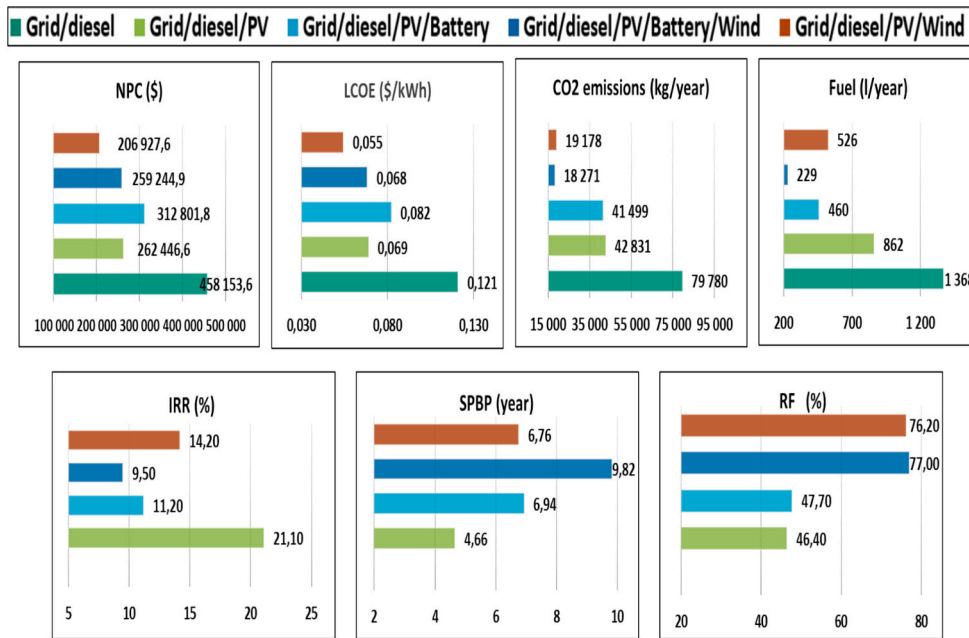


Fig. 39. Comparison of the techno-economic and environmental assessment of the different configurations for AGHAZARMADRAN.

- Policy and Regulatory Framework: Development of supportive policies and regulations that offer incentives for renewable energy investments, such as feed-in tariffs and tax breaks.
- Public-Private Partnerships: Collaborations between governments, private companies, and international organizations must be encouraged. Joint efforts can thus expedite renewable energy deployment and improve energy reliability.

- different companies such as mining industries that are found to be energy intensive and prevalent in Niger, while also taking into account their sizes including (small, medium, and large), and the variation in load profile curve shapes, which are influenced by the nature of the company's activities, and
- load profiles that are evaluated using captured dataset of several weeks of electrical load consumption for a more accurate load assessment. This work is limited by using captured dataset of one week for the load profile.

4.1. Suggestion for future work

Future studies should investigate the same work by considering,

CRedit authorship contribution statement

Maman Manirou Hachimou Jadi: Writing – original draft,

Validation, Formal analysis, Data curation. **Abdellah Benyoucef**: Writing – review & editing, Visualization, Supervision, Data curation. **Olayinka S. Ohunakin**: Writing – review & editing, Visualization, Supervision, Data curation, Conceptualization. **Isa Balogun**: Writing – original draft, Data curation. **Zulykhat Bukola Wasiu**: Writing – review & editing, Data curation. **Assarid Issaka Abdoukarim**: Writing – review & editing, Data curation. **Abdellatif Zerga**: Writing – review & editing.

Declaration of competing interest

The authors declare that they have no conflicting financial interests that could have appeared to hamper the work reported in this paper.

Data availability

All data utilized for this work are available upon request.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.est.2025.117435>.

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